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Far-field heat transfer and persistent heat
currents arising in a cylindrical non-reciprocal
configuration

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Abstract

This work investigates the consequences of breaking classical Kirchhoff's law of thermal radiation in a hollow, non-planar cylindrical configuration in the context of far-field heat transfer, where analytical solutions for the transmission coefficient are not feasible. To address this, we consider a partition of the cylinder into evenly spaced elements composed of both reciprocal and non-reciprocal materials. We develop a theoretical framework and a custom ray-tracing algorithm to perform the analysis. Our results show that non-reciprocity can lead to non-symmetric and non-zero net directional transmission coefficients between different elements of the cylinder at thermal equilibrium. Furthermore, we observe no persistent heat currents at thermal equilibrium within the interior of the cylinder for various realistic non-reciprocal material configurations. Our findings extend the results from established literature in both reciprocal and non-reciprocal regimes to a non-planar cylindrical configuration, without violating the general laws of thermodynamics.

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1 Introduction

Many photonic applications, for example, telecommunications and energy harvesting, can benefit from breaking Kirchhoff's law of thermal radiation, which states that a body's thermal emission equals its optical absorption. Violating this principle allows light to be re-directed into different directions than those from which it was absorbed, leading, for example, to the famous Landsberg limit of solar energy harvesting, which exceeds 90% [1]. Recent studies have demonstrated that this directional form of Kirchhoff's law of thermal radiation can be violated by applying strong magnetic fields to magneto-optical materials or by employing Weyl semimetals [2, 3, 4, 5]. This violation gives rise to a phenomenon defined as non-reciprocity where the heat flux density map is asymmetric with respect to the parallel wavevector. Materials that break Kirchhoff's law are called non-reciprocal and open new possibilities for controlling radiative energy flows, bringing thermal systems closer to their fundamental thermodynamic limits [6, 7, 8].

Until now, non-reciprocal heat transfer has been primarily investigated in simple planar geometries, with particular emphasis on the near-field regime [4, 9]. Studies have demonstrated a net directional transmission coefficient between two planar bodies [9] and persistent heat currents at thermal equilibrium in a symmetric three-nanoparticle configuration without violating the general laws of thermodynamics [3]. Building on these results, we advance the theory of non-reciprocity for non-planar structures by studying far-field heat transfer and the heat currents within the interior of a hollow cylindrical body. We develop a theoretical framework and a custom ray-tracing algorithm for this analysis. Our findings extend the results of [4, 9, 10] to a cylindrical configuration without violating the general laws of thermodynamics, and, additionally, demonstrate that no persistent heat currents arise inside the cavity, in contrast to the findings reported in [3].

This work is structured as follows: Section 2 presents the theoretical foundations of non-reciprocal heat transfer. Section 3 reviews the current state of the art and introduces the specific problem addressed in this work. Finally, Section 4 presents simulation results along with original findings.

2 Non-reciprocal heat transfer theory

In this section, we briefly introduce Kirchhoff's law of thermal radiation and present the theoretical framework leading to a more general law, known as the generalized Kirchhoff's law. Using this generalized law, we classify materials and extract their emission and absorption profiles.

All bodies radiate at a non-zero temperature. A blackbody is an idealized, perfectly opaque object representing the maximum possible emission and absorption profile and serves as the reference for defining the emissivity and absorptivity of real materials, as detailed in Appendix A.1, A.2.

2.1 Kirchhoff's law of thermal radiation

The general form of spectral Kirchhoff's law (1859) [11] states that, for a given polarization $\hat{p}$ and frequency ω , the emissivity ϵ in a direction $\hat{n}$ must equal the absorptivity α for light incident from the same direction:

$$\epsilon(\omega, \hat{n}, \hat{p}) = \alpha(\omega, -\hat{n}, \hat{p}^*) \quad (1)$$

Here, $\hat{p}^*$ denotes the complex conjugate¹ of the polarization vector $\hat{p}$.

Generally, the direction $\hat{n}$ is described in terms of the spherical angles (θ, φ) while $\hat{p}$ is the polarization state usually written in the in-plane (p) and out-of-plane (s) polarization basis. For unpolarized incoming light, Kirchhoff's law reduces to:

$$\epsilon(\omega, \theta, \varphi) = \alpha(\omega, \theta, \varphi)$$

While this law holds for most natural materials, recent studies have demonstrated that, under specific conditions, it is possible to violate spectral Kirchhoff's law [8, 12, 13], thereby motivating the search for a more general formulation, known as the Generalized Kirchhoff's law. It is important to note that this does not contradict the general laws of thermodynamics, since upon taking into account radiation emitted and absorbed towards and from all directions, the equality between overall emission ϵ and overall absorption α is restored, as detailed in Appendix A.2.

2.2 Generalized Kirchhoff's law of thermal radiation

To obtain the generalized Kirchhoff's law, we first introduce the transfer matrix formalism and the adjoint system to rigorously derive the emissivity and absorptivity profiles of materials and their adjoints.

Absorptivities and emissivities via the transfer matrix method

The transfer matrix method [14] is a formalism used to calculate the amplitudes of incident, reflected, and transmitted fields at a planar slab and, hence, obtain the emission, absorption and reflection profiles of materials. Applying electromagnetic theory yields the following 2×2 matrices, which relate the amplitudes of the incident field in the s - p polarization basis (a_s, a_p) to those of the reflected (r_s, r_p) and transmitted (t_s, t_p) fields in the same basis:

$$\begin{bmatrix} r_s \\ r_p \end{bmatrix} = \underbrace{\begin{bmatrix} r_{ss} & r_{sp} \\ r_{ps} & r_{pp} \end{bmatrix}}_{\mathbf{R} \text{ (Reflectivity matrix)}} \begin{bmatrix} a_s \\ a_p \end{bmatrix} \quad \begin{bmatrix} t_s \\ t_p \end{bmatrix} = \underbrace{\begin{bmatrix} t_{ss} & t_{sp} \\ t_{ps} & t_{pp} \end{bmatrix}}_{\mathbf{T} \text{ (Transmittivity matrix)}} \begin{bmatrix} a_s \\ a_p \end{bmatrix} \quad (2)$$

For opaque materials, the absorptivity and emissivity defined in Eq. (1) can be derived from the reflectivity matrix. Specifically, for a state $|\omega, \hat{n}, \hat{p}\rangle$ in bra-ket notation, commonly a two-component vector, the absorptivity (α) and emissivity (ϵ) are given by [15]:

$$\alpha(\omega, \hat{n}, \hat{p}) = 1 - \langle \omega, \hat{n}, \hat{p} | \mathbf{R}^\dagger \mathbf{R} | \omega, \hat{n}, \hat{p} \rangle, \quad \mathcal{A} := \mathbf{I} - \mathbf{R}^\dagger \mathbf{R} \quad (\text{Absorptivity matrix}) \quad (3a)$$

$$\epsilon(\omega, \hat{n}, \hat{p}) = 1 - \langle \omega, \hat{n}, \hat{p} | \mathbf{R} \mathbf{R}^\dagger | \omega, \hat{n}, \hat{p} \rangle, \quad \mathcal{E} := \mathbf{I} - \mathbf{R} \mathbf{R}^\dagger \quad (\text{Emissivity matrix}). \quad (3b)$$

A detailed derivation of Eq. (2), and Eq. (3) is found in Appendix B.1 and Appendix B.2 respectively.

¹For example, if $\hat{p}$ represents right circularly polarized light, then $\hat{p}^*$ corresponds to left circularly polarized light.

Adjoint systems and materials

By using the constitutive relations and Maxwell's equations, every material can be fully characterized by a 6×6 constitutive matrix, commonly referred $C(\omega, \mathbf{r})$, which relates the electric and magnetic fields. Specifically, the electric field $\mathbf{E}$ and magnetic field $\mathbf{H}$ are related to the electric displacement field $\mathbf{D}$ and the magnetic flux density $\mathbf{B}$ via:

$$\begin{pmatrix} \mathbf{D} \\ \mathbf{B} \end{pmatrix} = C(\omega, \mathbf{r}) \begin{pmatrix} \mathbf{E} \\ \mathbf{H} \end{pmatrix} = \begin{pmatrix} \epsilon(\omega, \mathbf{r}) & \zeta(\omega, \mathbf{r}) \\ \eta(\omega, \mathbf{r}) & \mu(\omega, \mathbf{r}) \end{pmatrix} \begin{pmatrix} \mathbf{E} \\ \mathbf{H} \end{pmatrix} \quad (4)$$

Each cell of the constitutive matrix represents a 3×3 submatrix or tensor. The symbols $\epsilon, \mu, \zeta, \eta$ are respectively the electric permittivity, magnetic permeability, electric-magnetic coupling strength, and magnetoelectric coupling strength. $\mathbf{r}$ denotes the spatial coordinates inside the material. A detailed explanation is given in Appendix A.3.

Given a system, we define it's **adjoint system** [2] by:

$$\tilde{C}(\omega, \mathbf{r}) = \begin{pmatrix} \epsilon^T(\omega, \mathbf{r}) & -\eta^T(\omega, \mathbf{r}) \\ -\zeta^T(\omega, \mathbf{r}) & \mu^T(\omega, \mathbf{r}) \end{pmatrix} \quad (5)$$

And we define the adjoint transformation to be:

$$\mathcal{T} : C(\omega, \mathbf{r}) \rightarrow \tilde{C}(\omega, \mathbf{r})$$

Note that $\mathcal{T}$ is involutory, i.e., the adjoint of the adjoint system is the original system. Although $\mathcal{T}$ is primarily a mathematical construction, it can have physical significance. For example, for magnetoptical materials, the adjoint system is equivalent to reversing the magnetic field.

The general reciprocity theorem leads to the constraint that the dyadic Green's function of mutually adjoint systems is mutually transpose. As a consequence the following relation between the reflectivity matrix of the original system $\mathbf{R}$ and the one of the adjoint system $\tilde{\mathbf{R}}$ holds:

$$\langle \omega, \hat{n}', \hat{p}' | \mathbf{R} | \omega, -\hat{n}, \hat{p} \rangle = \langle \omega, \hat{n}, \hat{p}^* | \tilde{\mathbf{R}} | \omega, -\hat{n}', \hat{p}'^* \rangle \quad (6)$$

A detailed derivation of the general reciprocity theorem is provided in Appendix A.5. A detailed explanation of the dyadic Green's function and derivation of Eq. (6) is provided in Appendix A.6.

Generalized Kirchhoff's law

We derive the generalized (adjoint) Kirchhoff's law [2], which extends its validity to arbitrary materials described by the constitutive matrix in Eq. (5). We begin with the full formal derivation, followed by detailed explanations of each step. Starting from an initial state defined by $|\omega, -\hat{n}, \hat{p}\rangle$, we obtain the following chain of equalities:

$$\begin{aligned} \alpha(\omega, -\hat{n}, \hat{p}) &\stackrel{(1)}{=} 1 - \langle \omega, -\hat{n}, \hat{p} | \mathbf{R}^\dagger \mathbf{R} | \omega, -\hat{n}, \hat{p} \rangle \\ &\stackrel{(2)}{=} 1 - \sum_{\sigma' \in \{s, p\}} \int d\hat{n}' \langle \omega, -\hat{n}, \hat{p} | \mathbf{R}^\dagger | \omega, \hat{n}', \sigma' \rangle \langle \omega, \hat{n}', \sigma' | \mathbf{R} | \omega, -\hat{n}, \hat{p} \rangle \\ &\stackrel{(3)}{=} 1 - \sum_{\sigma' \in \{s, p\}} \int d\hat{n}' \langle \omega, -\hat{n}', \sigma' | \tilde{\mathbf{R}}^\dagger | \omega, \hat{n}, \hat{p}^* \rangle \langle \omega, \hat{n}, \hat{p}^* | \tilde{\mathbf{R}} | \omega, -\hat{n}', \sigma' \rangle \\ &\stackrel{(4)}{=} 1 - \sum_{\sigma' \in \{s, p\}} \int d\hat{n}' \langle \omega, \hat{n}, \hat{p}^* | \tilde{\mathbf{R}} | \omega, -\hat{n}', \sigma' \rangle \langle \omega, -\hat{n}', \sigma' | \tilde{\mathbf{R}}^\dagger | \omega, \hat{n}, \hat{p}^* \rangle \\ &\stackrel{(2)}{=} 1 - \langle \omega, \hat{n}, \hat{p}^* | \tilde{\mathbf{R}} \tilde{\mathbf{R}}^\dagger | \omega, \hat{n}, \hat{p}^* \rangle \\ &\stackrel{(1)}{=} \tilde{\epsilon}(\omega, \hat{n}, \hat{p}^*). \end{aligned}$$

Where $\tilde{\epsilon}$ is the emissivity at the adjoint system. The equalities are satisfied due to:

1. Use of the definition of absorption and emission defined in Section 2.2 through the transfer matrix formalism.
2. Forward and backward insertion of the identity expansion in the basis $\{|\omega, \hat{n}', \sigma'\rangle\}$.
3. Use of the relation between the scattering matrices $\mathbf{R}$ and the scattering matrix of the adjoint system $\tilde{\mathbf{R}}$ defined in Eq. (6). Note that $|\sigma^*\rangle = |\sigma\rangle$.
4. Swapping the order of the inner products by transpose and conjugation.

Since the adjoint transformation $\mathcal{T}$ is involutory we also obtain a similar result for the pair emissivity and adjoint absorptivity. In conclusion:

Theorem 1 (Generalized (Adjoint) Kirchhoff's law). *Given an state $|\omega, \hat{n}, \hat{p}\rangle$, the emissivity and absorptivity of the original and adjoint system satisfy:*

$$\alpha(\omega, -\hat{n}, \hat{p}) = \tilde{\epsilon}(\omega, \hat{n}, \hat{p}^*) \quad \epsilon(\omega, \hat{n}, \hat{p}) = \tilde{\alpha}(\omega, -\hat{n}, \hat{p}^*).$$

Here, the $\tilde{\cdot}$ symbol means the respective adjoint system defined in Eq. (5).

2.3 Classification of materials and absorption-emission relations

We use the generalized Kirchhoff's law to classify materials into three groups and to establish their relationships between emissivity and absorptivity.

Let G be the group of all transformations (e.g., symmetries, rotations, translations) that leave the system invariant. Our aim is to identify materials for which $\mathcal{T} \in G$; in such cases, conventional Kirchhoff's law will hold. We define the subgroup $A \subset G$ as the set of all geometric transformations under which the system remains invariant². Additionally, let $\mathcal{T}B \subset G$, with B a set, denote the subset of compound transformations (geometric transformation followed by the adjoint transformation $\mathcal{T}$) under which the system remains invariant. The group

$$G = \{A, \mathcal{T}B\} \quad \mathcal{T}^2 = e \quad (\text{neutral element})$$

is mathematically termed as a Shubnikov point group. All Shubnikov groups can be classified into three types: *Gray group*, *Black and white group* and *Colourless group*. Each of these will provide a different conclusion for the pair of emissivity-absorptivity profile.

Reciprocal materials

Reciprocal materials are defined as those belonging to the gray group. For the gray groups, $B = A$, consequently, $\mathcal{T} = \mathcal{T}e \in G$ and since $\mathcal{T} \in G$, the system is invariant under the adjoint transformation and classical Kirchhoff's law holds³.

A nice consequence follows. From the mathematical derivation in Section 2.2, we observe that classical Kirchhoff's law is satisfied provided that

$$\mathbf{R} = \tilde{\mathbf{R}},$$

and, by using the reciprocity theorem provided in Appendix A.5 we observe the previous equation is satisfied if the constitute matrices of the original and adjoint system are equal:

²Rotations, reflections, symmetries, and improper rotations

³A direct consequence of $\mathcal{T} \in G$ is that the photonic band structure $\omega(k_{\parallel})$ is symmetric in the parallel wavevector space $\omega(k_{\parallel}) = \omega(-k_{\parallel})$. Hence, resonance frequencies for light incident at angles θ and $-\theta$ are identical, and Kirchhoff's law holds, as detailed in Appendix A.7

$$C(\omega, \mathbf{r}) = \begin{pmatrix} \varepsilon(\omega, \mathbf{r}) & \zeta(\omega, \mathbf{r}) \\ \eta(\omega, \mathbf{r}) & \mu(\omega, \mathbf{r}) \end{pmatrix} = \begin{pmatrix} \varepsilon^T(\omega, \mathbf{r}) & -\eta^T(\omega, \mathbf{r}) \\ -\zeta^T(\omega, \mathbf{r}) & \mu^T(\omega, \mathbf{r}) \end{pmatrix} = \tilde{C}(\omega, \mathbf{r}). \quad (8)$$

Consequently, if the dielectric and permittivity tensor are symmetric⁴ and there is no coupling between the fields ($\zeta = \eta = 0$), as is the case for most materials, Kirchhoff's law holds and the material is reciprocal⁵.

In such materials, the reflectivity and transmissivity matrices can be computed using Fresnel's equations, as detailed in Appendix B.6. The thermal relations are presented in Table 1. We use the definition of the reflectivity matrix from Eq. (2), and that for unpolarized light the total reflectivity is the average over *s*- and *p*-polarizations.

Non-reciprocal materials

Non-reciprocal materials are defined as those that do not belong to the gray group, that is, $\mathcal{T} \notin G$. As a result, these materials do not obey classical Kirchhoff's law. In that case, generalized Kirchhoff's law and Shubnikov group theory are used to classify non-reciprocal materials into two groups: The *Black and white* and *Colourless* groups [2].

Black and white materials In black and white materials, $B \neq \emptyset$ and $B \cap A = \emptyset$. In this case, $\mathcal{T} \notin G$ but since B is non-empty, G contains some compound symmetries leading to a set of modified Kirchhoff's law that relate specific pairs of emissivities and absorptivities.

A straightforward criterion for non-reciprocity is an asymmetric dielectric tensor, which violates the condition in Eq. (8). A standard approach to breaking symmetry in the photonic band structure is to apply a strong external magnetic field to certain materials. For a magnetic field in the *y*-direction (Voight configuration), the asymmetric dielectric tensor results in:

$$\epsilon_{NR} = \begin{pmatrix} \epsilon_{xx} & 0 & i\epsilon_a \\ 0 & \epsilon_{yy} & 0 \\ -i\epsilon_a & 0 & \epsilon_{zz} \end{pmatrix} \quad \mu = \mu_0 \quad \zeta = 0 \quad \eta = 0 \quad (9)$$

where $\epsilon_{ii} \in \mathbb{C}$ for $i = x, y, z$ and $\epsilon_a \in \mathbb{R}$. The non-zero imaginary values $i\epsilon_a$ result from applying a strong external field to the material [16, 12].

Relevant materials are:

1. *Magneto-optical materials*: Magneto-optical materials are materials that exhibit the dielectric tensor of Eq. (9) upon the application of a magnetic field along the direction where the optical activity exists. One example is an *n*-InAs grating subject to a magnetic field along the *x*-axis [12]. A field of $B = 3$ T has shown to break Kirchhoff's law, with the threshold reducible to $B = 0.3$ T.
2. *Weyl semimetals*: These materials break reciprocity and time-reversal symmetry without requiring an external magnetic field because of their topological features [5]. Their structure typically mirrors the magneto-optical configuration [17].

For these materials, compound symmetries in $\mathcal{TB} \subset G$ such as $\mathcal{T}\sigma_{xz}$, where σ_{xz} denotes reflection along the *xz*-axis are possible [2]. As a result, the general relation $\epsilon(\omega, \theta, \varphi) = \alpha(\omega, -\theta, \varphi)$ holds for any specular emitter [8, 18, 19].

⁴The spectral decomposition theorem implies that symmetric matrices have real eigenvalues, which leads to the representation of the optical index ellipsoid.

⁵Examples include vacuum, glass or silver.

The absorption profile for a black and white material with permittivity tensor given by Eq. (9) is obtained by computing the reflection coefficients, which is possible by solving the wave equation in the interface of the material for planar waves [4]

$$k^2 \mathbf{E} - \mathbf{k}(\mathbf{k} \cdot \mathbf{E}) = k_0^2 \epsilon_2 \mathbf{E}$$

where k_0 is the wavevector in the vacuum. A detailed explanation of the wave equation is given in Appendix A.4. The previous relation has to be satisfied for non-trivial incident amplitudes $\mathbf{E}$ leading to a generalized eigenvalue problem

$$\det(\mathbf{k}\mathbf{k}^T - k^2 \mathbf{I} - k_0^2 \epsilon_2) = 0. \quad (10)$$

An analytical solution of Eq. (10) is provided in [4]. For s-polarization, the material is reciprocal because it does not involve the ϵ_a term in Eq. (9) and Fresnel's equations are recovered. For p-polarization, the z-component of the wavevector inside the material, k_{z2} , is

$$k_{z2} = \pm \sqrt{\epsilon_v k_0^2 - k_x^2} \quad \epsilon_v \equiv \epsilon_{xx} - \epsilon_a^2 / \epsilon_{zz}.$$

Leading to a closed form for r_p given by

$$r_p = \frac{k_{z1} \epsilon_v - k_{z2} + i k_x \epsilon_a / \epsilon_d}{k_{z1} \epsilon_v + k_{z1} - i k_x \epsilon_a / \epsilon_d}. \quad (11)$$

Here, k_{z1} and k_x are the x, z components of the wavevector in medium 1. In this case, $|r_p(k_x)|^2 \neq |r_p(-k_x)|^2$, leading to an asymmetric photonic band structure and Kirchhoff's law is violated, as detailed in Appendix A.7. Note that larger values of ϵ_a do not necessarily lead to stronger non-reciprocal effects; indeed, in the limit $\epsilon_a \rightarrow \infty$, non-reciprocity vanishes. A summary of the relations obtained for specular black and white materials are presented in Table 1.

Colourless materials For colourless materials, $B = \emptyset$, hence, $\mathcal{T} \notin G$, and G does not contain compound symmetries. As a result, no relation exists between any directional emissivity and absorptivity. An example of a such material is given in Appendix B.5, where both the magnetoelectric coupling and magnetic permeability tensors are non-zero.

To construct colourless emitters, it is necessary to design diffractive, nonreciprocal structures that break the symmetry between emission and absorption by redirecting incoming rays into different channels or directions [17]. This approach offers greater flexibility in the design of nonreciprocal materials [8], however, it also introduces challenges for simulation. For colourless materials the generalized relations are provided in Table 1.

Material type	Emissivity / Absorptivity	Reflectivity
Reciprocal	$\epsilon(\theta) = \alpha(\theta), \quad \alpha(\theta) = 1 - R(\theta)$	$R(\theta) = \frac{ r_s ^2 + r_p ^2}{2}$
Black and White (Specular)	$\epsilon(\theta) = \alpha(-\theta), \quad \alpha(\theta) = 1 - R(\theta)$	$R(\theta) = \frac{ r_s ^2 + r_p ^2}{2}$
Colourless (Diffractive)	$\epsilon(\theta) : \text{Eq. (3a)}, \quad \alpha(\theta) : \text{Eq. (3b)}$	$R(\theta) = 1 - \alpha(\theta)$

Table 1: Summary of material classification and their thermal radiative exchange relations. For reciprocal materials, the standard procedure using Fresnel's equations is presented. In the black and white case, specular emission [17] leads to an asymmetric relation between emissivity and absorptivity, involving the reflection coefficient r_p in Eq. (11). For colourless materials, the relations follow the general guidelines detailed in Section 2.2. The value of φ is omitted for simplicity.

3 Underlying motivation and Problem statement

3.1 Overview of prior advances in non-reciprocal heat transfer

Motivated by recent advances in breaking Kirchhoff's law, specifically, the symmetry between emissivity and absorptivity, we highlight two foundational works on non-reciprocal heat transfer. First, Fan et al. [9] studied directional heat transfer between two parallel non-reciprocal planes and showed how breaking reciprocity leads to a non-zero net directional transmission coefficient. Second, Zhu and Fan [3] theoretically demonstrated persistent heat currents at thermal equilibrium in specific non-reciprocal systems. Both studies considered the thermal near-field, where the two objects exchanging heat are placed in close proximity with respect to the thermal wavelength.

Breaking the symmetry of the heat flux between two non-reciprocal planes

Fan et al. [9] consider two planar surfaces, semi-infinite along z and infinite in the xy -plane, separated by a vacuum gap d at the same temperature (T). To provide insight into their results, we reproduce a simplified far-field result in two dimensions (x, z), noting that their full derivation also covers the near-field regime. The goal is to analyze the directional transmission coefficient between planes 1 and 2, emphasizing asymmetric heat exchange due to non-reciprocal materials. To this end, an optics model is used, as illustrated in Figure 1.

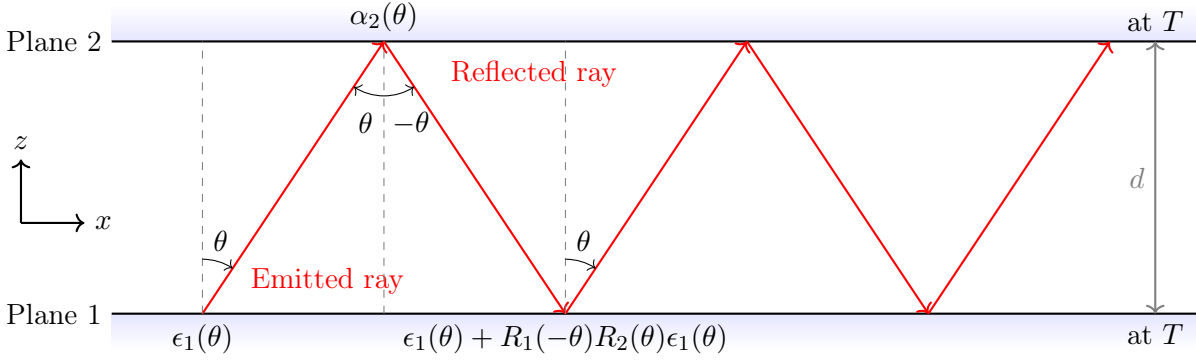


Figure 1: Scheme of the ray-tracing algorithm between two non-parallel planes, starting from the most left point and accounting for multiple reflections. The angle θ is measured relative to each plane's normal and determines the transmitted intensity along the ray's path using the emission ϵ , absorption α and reflection profiles R .

The spectral directional transmission coefficient from body 1 to body 2 in direction θ , denoted $S_{1 \rightarrow 2}$, is obtained by summing the contributions of all rays propagating in that direction:

$$S_{1 \rightarrow 2}(\theta) = \epsilon_1(\theta) \alpha_2(\theta) \cos(\theta) \sum_{n \geq 0} [R_1(-\theta) R_2(\theta)]^n$$

where $\epsilon_i(\theta)$, $\alpha_i(\theta)$, and $R_i(\theta)$ denote the emission, absorption, and reflection coefficients at direction θ for plane $i \in \{1, 2\}$. The factor $\cos(\theta)$ originates from Lambert's law and accounts for the directional distribution of emitted rays⁶.

The same analysis applies for direction $2 \rightarrow 1$. Using the geometric series property, we obtain:

$$S_{1 \rightarrow 2}(\theta) = \frac{\epsilon_1(\theta) \alpha_2(\theta)}{1 - R_1(-\theta) R_2(\theta)} \cos(\theta) \quad S_{2 \rightarrow 1}(\theta) = \frac{\epsilon_2(\theta) \alpha_1(\theta)}{1 - R_2(-\theta) R_1(\theta)} \cos(\theta)$$

The angle convention θ in $S_{1 \rightarrow 2}(\theta)$ is measured relative to the normal of plane 1, while in $S_{2 \rightarrow 1}(\theta)$, it is measured relative to the normal of plane 2, see Figure 1.

⁶This factor arises naturally when changing variables from wavevector to angular coordinates, as detailed in Section 3.2.2.

When bodies are made of reciprocal materials, it leads to symmetric absorptivity and emissivity profiles, resulting in balanced directional heat transfer:

$$\epsilon_i(\theta) = \alpha_i(\theta) \quad i \in \{1, 2\} \quad \implies \quad S_{1 \rightarrow 2}(\theta) = S_{2 \rightarrow 1}(\theta)$$

Considering a specular black-and-white non-reciprocal material in plane 1 and a reciprocal material in plane 2 a net directional transmission coefficient is obtained.

$$\epsilon_1(\theta) = \alpha_1(\theta) \quad \epsilon_2(\theta) = \alpha_2(-\theta) \quad \implies \quad S_{1 \rightarrow 2}(\theta) \neq S_{2 \rightarrow 1}(\theta)$$

As expected, Fan et al. [9] showed that the second law of thermodynamics is not violated, i.e. that there is no heat flux between the two bodies when they are at thermal equilibrium. Integrating over all angles, the heat flux from body 1 to body 2 balances the one from body 2 to body 1 at each frequency. Numerical simulations of two planar surfaces, composed of different reciprocal/non-reciprocal materials confirm these results for various wavelengths and separation gap between the bodies.

Persistent heat currents arising in non-reciprocal configurations

Zhu and Fan [3] demonstrated near-field persistent heat currents at thermal equilibrium in non-reciprocal configurations involving at least three bodies. They proposed three spheres arranged in a triangular symmetry, each with a submicron separation and diameter scale, as illustrated in Figure 2. The spheres are made of a magneto-optical material. At thermal equilibrium, no persistent heat currents are observed when the materials are reciprocal. However, under a strong magnetic field ($B = 3$ T), the materials become non-reciprocal, and theory predicts persistent heat currents circulating clockwise.

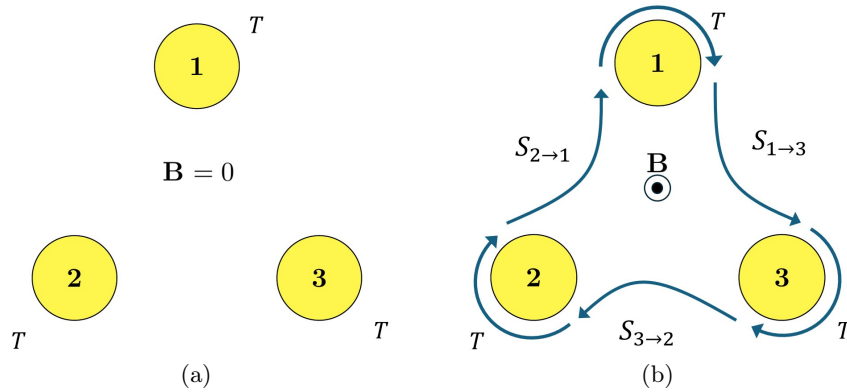


Figure 2: A strong magnetic field is applied to a three-body magneto-optical configuration. (a), the system is at equilibrium without an external magnetic field. In (b), the magnetic field breaks reciprocity, inducing a net directional transmission coefficient between elements ($S_{i \rightarrow j}$), which gives rise to persistent heat currents at thermal equilibrium, illustrated by the blue arrows (not scaled).

This behaviour is observed by computing the Poynting vector direction while neglecting far-field heat transfer (several orders of magnitude smaller). The study does not violate the second law of thermodynamics, as the net heat emitted and absorbed by each sphere is zero, i.e., $S_{1 \rightarrow 3} = S_{3 \rightarrow 2} = S_{2 \rightarrow 1}$. Further investigations were carried out in 2018, extending the study to systems of many spheres under non-equilibrium conditions [20]. In this case, a configuration with one sphere at 300 K and the others at 200 K exhibited a non-uniform asymmetric heating response. More recently, Biehs et al. (2025) theoretically proved that the Poynting vector is divergence-free at thermal equilibrium reinforcing consistency with the second law [21].

3.2 Formulation of non-reciprocal heat transfer in a cylindrical geometry

We extend the study of non-reciprocal heat transfer by examining its behaviour in the far-field using the simplest non-planar geometry.

Consider a hollow infinite cylinder embedded in vacuum defined by the following parametrization:

$$\psi : (\alpha, z) \in [0, 2\pi) \times \mathbb{R} \longrightarrow (R \cos(\alpha), R \sin(\alpha), z) \in \mathbb{R}^3$$

And splitted into adjacent elements Ω_i , $i \in \{0, 1, \dots, n-1\}$ defined by:

$$\Omega_i = \psi \left(\left[\frac{2\pi}{n}i, \frac{2\pi}{n}(i+1) \right), \mathbb{R} \right) \quad i \in \{0, \dots, n-1\} \quad (12)$$

Suppose $R \gg \lambda$ where λ is the thermal wavelength (neglecting near-field heat transfer), and we consider the surface opaque (neglecting transmission). Each element Ω_i is assigned either a reciprocal or non-reciprocal material, as illustrated in Figure 3. Our study aims to determine the **transmission coefficients** between any two elements Ω_i and Ω_j , denoted by $S_{i \rightarrow j}$, to analyze the **assymetry in heat transfer**, and to investigate the possibility of **persistent heat currents** inside the cavity.

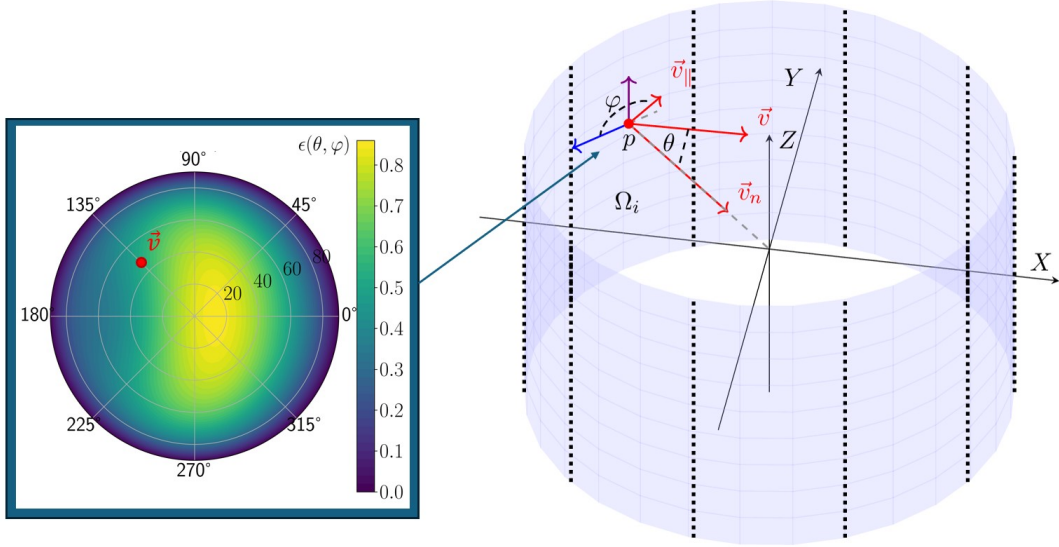


Figure 3: Cylindrical configuration: the element Ω_i , defined in Eq. (12), is shown. A point p emits in direction $\vec{v}$ decomposed into in-plane ($v_{||}$) and out-of-plane (v_n) components, and define the emission direction (θ, φ) used in the emissivity profile.

We adopt a similar mathematical framework as in Section 3.1. Specifically, we build upon the theory developed in Section 2 and adapt the method in [9] to our cylindrical configuration. In the following section, we derive the transmission coefficient expression, primarily following [12] and Zhang's textbook [22].

3.2.1 Mathematical formulation of the heat transfer coefficient

The heat absorbed by an element Ω_i at a given time t , denoted $\dot{Q}(t)$, can be computed as the flux of the Poynting vector $\mathbf{S}$ across its surface:

$$\dot{Q}(t) = \int_{\Omega_i} q(\mathbf{r}, t) \cdot d\Omega_i = \int_{\Omega_i} \mathbf{S}(\mathbf{r}, t) \cdot \mathbf{n} d\Omega_i$$

Where $q(\mathbf{r}, t)$ is the heat density transmitted and $\mathbf{r}$ is a point of the cylinder. The Poynting vector is derived from the conservation of energy theorem detailed in Section A.8:

$$\mathbf{S}(\mathbf{r}, t) = \mathbf{E}(\mathbf{r}, t) \times \mathbf{H}(\mathbf{r}, t)$$

Of most interest is to compute the time-averaged value of the heat flux over a long period of time:

$$Q = \left\langle \int_{\Omega_i} \mathbf{S}(\mathbf{r}, t) \cdot \mathbf{n} d\Omega_i \right\rangle = \int_{\Omega_i} \langle \mathbf{S}(\mathbf{r}, t) \cdot \mathbf{n} \rangle d\Omega_i \quad (13)$$

The following chain of equalities rewrites the integrand of Eq. 13 into a more manageable expression. Here, we provide an intuitive explanation of each step; a full analytical derivation is given in Appendix B.3. For a point at position $\mathbf{r}$ on the cylinder, the average transmitted heat is:

$$\langle \mathbf{S}(\mathbf{r}, t) \cdot \mathbf{n} \rangle =^{(1)} \frac{1}{2} \text{Re} \int_0^\infty d\omega \int \frac{d\mathbf{k}_\parallel}{(2\pi)^4} \mathbf{n} \cdot [\mathbf{E}(\mathbf{k}_\parallel, z, \omega) \times \mathbf{H}^*(\mathbf{k}_\parallel, z, \omega)] \quad (14a)$$

$$=^{(2)} \text{Tr} \frac{1}{2} \int_0^\infty d\omega \int \frac{d\mathbf{k}_\parallel}{(2\pi)^4} \text{Re} \left[(\mathbf{E}_i + \mathbf{E}_r) [Z^{-1}(\mathbf{E}_i - \mathbf{E}_r)]^\dagger \right] \quad (14b)$$

$$=^{(3)} \text{Tr} \frac{1}{2} \int_0^\infty d\omega \int \frac{d\mathbf{k}_\parallel}{(2\pi)^4} \left[(\mathbf{1} - \mathbf{R}^\dagger \mathbf{R}) Z^{-1} \mathbf{E}_i \mathbf{E}_i^\dagger \right] \quad (14c)$$

Where $\mathbf{E}_i$, $\mathbf{E}_r$ are the incoming and reflected fields in the (s)-(p) polarization basis at point $\mathbf{r}$ and $\mathbf{R}$, Z are the reflectivity matrix and the impedance matrix.

In equality (1), a Fourier transform is applied both in time ($t \rightarrow \omega$) and position ($\mathbf{r}_\parallel \rightarrow \mathbf{k}_\parallel$). This removes the time average, restricts the analysis to each frequency ω , and expresses vectors in terms of $\mathbf{k}_\parallel$, which simplifies computing reflectivity coefficients. In equality (2), Maxwell's equations rewrite the cross product of electric and magnetic fields as products of incoming and outgoing fields. Using matrices and traces is convenient here as it enables sums of (s) and (p) polarizations. In equality (3), properties of matrix traces and real-valued functions are applied to rewrite the previous equation.

We now consider the incoming field at a given point inside the cylinder. It consists of two main contributions: direct emission from the surface element facing the point, and the effect of multiple reflections from other elements contributing to the same direction. This is expressed as:

$$\mathbf{E}_i = \mathbf{E}_1 e^{i\varphi_1} + \mathbf{R}_1 \mathbf{E}_2 e^{i(\varphi_1 + \varphi_2)} + \mathbf{R}_1 \mathbf{R}_2 \mathbf{E}_3 e^{i(\varphi_1 + \varphi_2 + \varphi_3)} + \mathbf{R}_1 \mathbf{R}_2 \mathbf{R}_3 \mathbf{E}_4 e^{i(\varphi_1 + \varphi_2 + \varphi_3 + \varphi_4)} + \dots$$

Here, the subscripts refer to different points along the ray path. Since the direction is fixed, the contribution from point 2 arrives via a reflection at point 1, and the exponential terms $e^{i\varphi_j}$ account for the phase shifts introduced by propagation through the material. A similar process applies to all other points, with successive reflections and phase accumulations.

Next, we relate the fluctuations of the electric field inside the cylinder to the emission rate of the emitting body by using the fluctuation-dissipation theorem [23]. For far-field heat transfer, it leads to the intensity of the emitted field $\mathbf{E}_j$:

$$\langle \mathbf{E}_j \mathbf{E}_j^\dagger \rangle = (2\pi)^2 \frac{Z}{\pi} \Theta(\omega, T) \underbrace{\left[\mathbf{I} - \mathbf{R}_j \mathbf{R}_j^\dagger \right]}_{\text{Emissivity matrix } \mathcal{E}_j} \quad \text{for } j \in \{1, 2, 3, \dots\}$$

Where $\Theta(\omega, T)$ is the mean photon energy, detailed in the end of Appendix B.3. Assuming that fields emitting from different points $\mathbf{E}_i, \mathbf{E}_j$ are uncorrelated and using the definitions (3a), (3b) of absorptivity, emissivity matrix, Eq (14c) is simplified to:

$$\langle \mathbf{S}(\mathbf{r}, t) \cdot \mathbf{n} \rangle = \text{Tr} \frac{1}{2} \int_0^\infty d\omega \int \frac{d\mathbf{k}_\parallel}{(2\pi)^4} \left[(\mathbf{1} - \mathbf{R}^\dagger \mathbf{R}) Z^{-1} (\langle \mathbf{E}_1 \mathbf{E}_1^\dagger \rangle + \mathbf{R}_1 \langle \mathbf{E}_2 \mathbf{E}_2^\dagger \rangle \mathbf{R}_1^\dagger + \dots) \right] \quad (15a)$$

$$= \text{Tr} \int_0^\infty \frac{d\omega}{2\pi} \int \frac{d\mathbf{k}_\parallel}{(2\pi)^2} \Theta(\omega, T) \mathcal{A} \left(\mathcal{E}_1 + \mathbf{R}_1 \mathcal{E}_2 \mathbf{R}_1^\dagger + \mathbf{R}_1 \mathbf{R}_2 \mathcal{E}_3 \mathbf{R}_2^\dagger \mathbf{R}_1^\dagger + \dots \right) \quad (15b)$$

We considered a simplification by replacing the absorptivity and emissivity matrices with a weighted average of the emission and absorption profiles over s- and p-polarizations. The expression is rewritten as:

$$\langle \mathbf{S}(\mathbf{r}, t) \cdot \mathbf{n} \rangle = 2 \int_0^\infty \frac{d\omega}{2\pi} \int \frac{d\mathbf{k}_\parallel}{(2\pi)^2} \Theta(\omega, T) S(\mathbf{r}, \mathbf{k}_\parallel, \omega) \quad (16a)$$

$$S(\mathbf{r}, \mathbf{k}_\parallel, \omega) = \alpha(\mathbf{r}, \mathbf{k}_\parallel, \omega) (\epsilon_1 + R_1 \epsilon_2 + R_1 R_2 \epsilon_3 + \dots) \quad (16b)$$

All relevant information is captured by the transmission coefficient $S(\mathbf{r}, \mathbf{k}_{\parallel}, \omega)$ at point $\mathbf{r}$ ⁷. Note that setting $S(\mathbf{r}, \mathbf{k}_{\parallel}, \omega) = 1$ recovers Planck's law:

$$\langle \mathbf{S}(\mathbf{r}, t) \cdot \mathbf{n} \rangle = 2 \int_0^\infty \frac{d\omega}{2\pi} \int \frac{d\mathbf{k}_{\parallel}}{(2\pi)^2} \Theta(\omega, T) = \sigma T^4 \quad (17)$$

3.2.2 Numerical computation of the transmission coefficient

We generalize the transmission coefficient expression at a point on the cylinder $S(\mathbf{r}, \mathbf{k}_{\parallel}, \omega)$ to the transmission coefficient between elements $S_{i \rightarrow j}$. Combining Eqs. (13) and (16), the heat transmitted to an element Ω_j of the cylinder is:

$$Q = \int_{\Omega_i} \left[2 \int_0^\infty \frac{d\omega}{2\pi} \int \frac{d\mathbf{k}_{\parallel}}{(2\pi)^2} \Theta(\omega, T) S(\mathbf{r}, \mathbf{k}_{\parallel}, \omega) \right] d\Omega_i$$

Where $S(\mathbf{r}, \mathbf{k}_{\parallel}, \omega)$ is the spectral directional transmission coefficient. That is, the directional heat that is transmitted into point $\mathbf{r} := [\alpha, z]$ at frequency ω in proportion to the heat emitted by a blackbody.

We define the spectral transmission coefficient from a point $p := [\alpha, z]$ on the cylinder to an element Ω_i , in the direction $\mathbf{k}_{\parallel}$, as a mapping

$$S_i(\mathbf{p}, \mathbf{k}_{\parallel}, \omega) : \mathbb{R}^2 \times \mathbb{R}^2 \times \mathbb{R} \rightarrow [0, 1]$$

satisfying that the radiative heat density transferred from point $\mathbf{p}$ to the element Ω_i is:

$$\langle \mathbf{S}_{p \rightarrow i} \rangle = \int_0^\infty \frac{d\omega}{2\pi} \int \frac{d\mathbf{k}_{\parallel}}{(2\pi)^2} \Theta(\omega, T) S_i(\mathbf{p}, \mathbf{k}_{\parallel}, \omega)$$

To define the average directional spectral transmission coefficient normalized by blackbody emission, we introduce the spectral transmission coefficient from point $\mathbf{p}$ to element Ω_i as:

$$S_i(\mathbf{p}, \omega) = \frac{1}{C_1} \int d\mathbf{k}_{\parallel} S_i(\mathbf{p}, \mathbf{k}_{\parallel}, \omega) \quad C_1 = \int d\mathbf{k}_{\parallel} = \int_0^{\frac{\omega}{c}} \int_0^{2\pi} r dr d\varphi = \pi \left(\frac{\omega}{c} \right)^2$$

Finally, we provide the simulated quantity, expressing the transmission coefficient over an element by integrating across a section of the cylinder:

Definition 1 (Monochromatic transmission coefficient). *Let $p := [\alpha, z] \in \Omega_i$. We define the monochromatic transmission coefficient from element i to element j as:*

$$S_{i \rightarrow j}(\omega) = \frac{1}{C_2} \int_{\frac{2\pi R}{n} i}^{\frac{2\pi R}{n} (i+1)} S_j(\mathbf{p} = [\alpha, z], \omega) d\alpha \quad C_2 = \int_{\frac{2\pi R}{n} i}^{\frac{2\pi R}{n} (i+1)} d\alpha = \frac{2\pi R}{n}$$

By integrating spectrally over all frequencies we obtain that the radiation density transmitted from element Ω_i to element Ω_j is:

$$\langle \mathbf{S}_{i \rightarrow j} \rangle dz = \frac{2\pi R}{n} \int_0^\infty \frac{w^2 dw}{4\pi^2 c^2} \Theta(w, T) S_{i \rightarrow j}(\omega) = \frac{2\pi R}{n} (S_{i \rightarrow j} \sigma T^4) dz$$

Where $S_{i \rightarrow j}$ is the transmission coefficient between elements.

For simulations, it is convenient to express the transmission coefficient in spherical coordinates (θ, φ) . This is done by changing variables from $\mathbf{k}_{\parallel}$ to (θ, φ) and transforming the integration domain accordingly. We obtain:

$$S(\mathbf{p}, \mathbf{k}_{\parallel}, \omega) d\mathbf{k}_{\parallel} = \left(\frac{\omega}{c} \right)^2 \cos \theta \sin \theta S(\mathbf{p}, \theta, \varphi, \omega) d\theta d\varphi \quad (18)$$

When computing the transmission coefficient under the angle transformation a cosine factor arises known as Lambert's law.

⁷This derivation considers only propagating waves as evanescent waves only contribute when R is of the order of the thermal wavelength.

3.2.3 A specular ray tracing algorithm

In this work, we present an in-house specular ray tracing algorithm specifically designed for our cylindrical configuration⁸. Our approach builds on Eq. (16) as a foundational pillar. This method has proven effective [8] for materials exhibiting specular reflection (reciprocal and specular black-and-white materials). For colourless materials, multiple diffraction channels may be relevant [17], making specular ray tracing a non-representative method. Consequently, we limit our study to materials that reflect specularly.

It is worth noting that standard ray-tracing algorithms may account for constructive or destructive interference effects, such as those in a Fabry–Pérot interferometers. However, this omission is justified here, since interference typically requires perfectly coherent light in close configurations. When the reflective surfaces in a Fabry–Pérot cavity are sufficiently separated or light is incoherent, interference effects vanish, supporting the validity of our ray-based approach.

The following code has two main features, computing the transmission coefficient between elements, and simulating the monochromatic heat currents arising inside the cylinder:

Numerical method for the transmission coefficient ($\mathbf{S}_{i \rightarrow j}$) The proposed algorithm discretizes each element into a mesh of equidistant points. A specular ray tracing algorithm is applied for points only at $z = 0$ due to the uniformity in the z -direction. The step-by-step guide follows:

1. Initialize a partition of the cylinder into equidistant elements Ω_i , as defined in Eq. (12), assigning emission, absorption, and reflection coefficients at each element, as illustrated in Figure 3.
2. Discretize each element Ω_i into a fine mesh, where each mesh cell is approximated by small rectangular area dA_1 . For a given direction $\mathbf{k}_{\parallel}$ and starting point dA_1 , run the ray-tracing algorithm until the ray intensity drops below a predefined tolerance due to absorption. The transmitted intensity to each intersected element is recorded accordingly.
3. Compute the total directional transmission coefficient by applying the previous formulas over all surface elements dA_j at $z = 0$. Important factors include the number of rays emitted and the influence of distance between elements. One can follow Eq. (18) for integration, thus, for each element dA_1 , rays are emitted uniformly in angular space. Another way of reasoning is presented in Appendix B.4.
4. We perform this measurement for each cell mesh dA_1 of each element at $z = 0$ and we normalize the quantities as defined in Section 3.2.2.

Numerical method for the heat flux vector $\mathbf{v}(\mathbf{p}, \omega)$ The proposed algorithm simulates the heat flux vector in two regions: The points inside the hollow cylinder, and the contour heat flux in the boundary of the cylinder.

Inner heat flux

1. For a point $\mathbf{p}$ inside the cylinder (not on the surface, either absorbing or emitting), define the average directional heat transmission coefficient as:

$$\mathbf{v}(\mathbf{p}, \omega) = \frac{1}{\pi} \int_0^\pi \int_0^{2\pi} \tilde{S}(\mathbf{p}, \theta, \varphi, \omega) (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta) d\varphi d\theta, \quad (19)$$

where $\mathbf{v}(\mathbf{p}, \omega)$ gives the vector of the heat flux at point $\mathbf{p}$, and $\tilde{S}(\mathbf{p}, \theta, \varphi, \omega)$ is the transmission coefficient excluding the absorption at $\mathbf{p}$. Here, we followed a similar reasoning to previous result, in such case, the cosine factor of Lambert’s law is not necessary.

⁸An alternative method involves a finite element approach, where the cylinder is divided into many small segments, and boundary conditions along with the radiance of each element are incorporated to form a system of equations and obtain a solution at thermal equilibrium.

2. Repeat this over a mesh of points $\mathbf{p}$ inside the cylinder to form a vector field. We normalize Eq. (19) by π to ensure:

$$Q = \sigma T^4 \int_A \mathbf{v}(\mathbf{p}, \omega) \cdot \mathbf{n} dA$$

where Q is the heat flux through surface A with normal component $\mathbf{n}$.

Contour heat flux

1. For a point $\mathbf{p} := [\alpha, 0]$ of the cylinder the integral is split into two hemispherical components: one for outgoing heat (emission) and one for incoming heat (transmission coefficient):

$$\mathbf{v}(\mathbf{p}, \omega) = \frac{1}{\pi} \int_0^{\pi/2} \int_0^{2\pi} (\epsilon(\omega, \varphi, \theta) - S(\mathbf{p}, \omega, \varphi)) (\cos \varphi \sin \theta, \sin \varphi \sin \theta, \cos \theta) d\theta d\varphi$$

where $S(\mathbf{p}, \omega, \varphi)$ is the transmission coefficient, and $\epsilon(\omega, \varphi)$ is the emissivity profile at point $\mathbf{p}$ in local coordinates as shown in Figure 3.

2. Repeat for equidistant points along the contour. Similarly to the inner heat, we normalize by π so that:

$$Q = \sigma T^4 \int_A \mathbf{v} \cdot \mathbf{n} dA.$$

4 Numerical Simulations and Results

We present the results for the cylindrical configuration defined Section 3.2 using the quantities from Section 3.2.2 and the algorithms from Section 3.2.3. We compare them with prior literature reviewed in Section 3.1.

4.1 Monochromatic transmission coefficient

Monochromatic transmission coefficient for a single-material cylinder

Numerical simulations of the monochromatic transmission coefficient are performed for a cylindrical configuration composed of a single material, following the formalism in Section 3.2.2. We first consider an ideal, reciprocal blackbody cylinder. Next, we simulate two non-reciprocal materials: (1) a Weyl semimetal with a permittivity tensor and emission-absorption properties detailed in Section 2.3, and (2) an idealized non-reciprocal emitter, termed *On-off*, which fully absorbs from the left hemisphere and fully emits from the right hemisphere. The directional emission profiles in spherical coordinates for each material are shown in Figure 4.

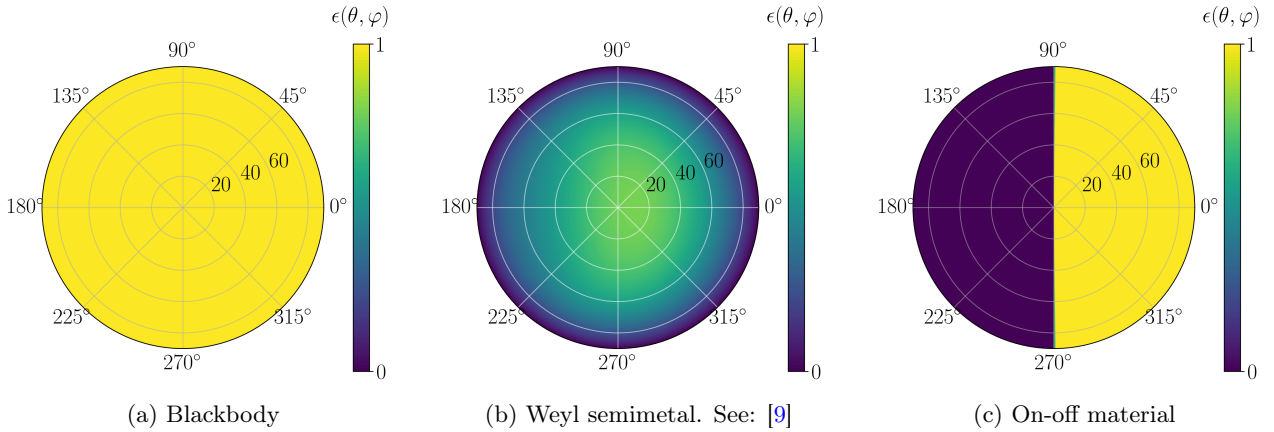


Figure 4: Polar plots of the directional emission profiles as a function of the azimuthal and polar angles (θ, φ) . (a) a reciprocal blackbody emitter with uniform unit emission, (b) a non-reciprocal Weyl semimetal, as described in Section 2.3, (c) an idealized non-reciprocal on-off material. The two non-reciprocal emitters show asymmetric profile with enhanced emission for $\cos(\varphi) \geq 0$.

The cylinder is partitioned into $n = 8$ elements. To illustrate the transmission coefficients $S_{i \rightarrow j}$, a graph is drawn in which arrows represent the net transmission coefficient $H_{i \rightarrow j} = S_{i \rightarrow j} - S_{j \rightarrow i}$ between any two elements. Results are presented in Figure 5.

The net transmission coefficient between surfaces, $H_{i \rightarrow j}$, characterizes the monochromatic heat flow between elements. For reciprocal materials, the emission profile is symmetric, and we expect equal heat exchange for positive and negative parallel wavevectors. As a result, the net heat exchange between any pair of elements vanishes, i.e., $S_{i \rightarrow j} = S_{j \rightarrow i}$, consistent with the observations of Fan et al. [9]. Simulations involving other reciprocal materials, such as aluminium or silver, exhibit the same symmetry. In contrast, non-reciprocal materials exhibit asymmetric emission profiles, leading to predominant emission in a preferred direction (e.g., towards the right hemisphere). Since this behaviour is repeated for each element, the resulting net transmission coefficient between elements is non-zero, enabling directional heat channels and extending the results from [9] to non-planar surfaces. The effect is amplified in the idealized on-off emitter, which shows net monochromatic transmission coefficients of an order of magnitude larger. The symmetry observed in the transmission graph reveals a circular heat flow pattern, driven by preferential absorption-emission channels, similar to the findings of [3].

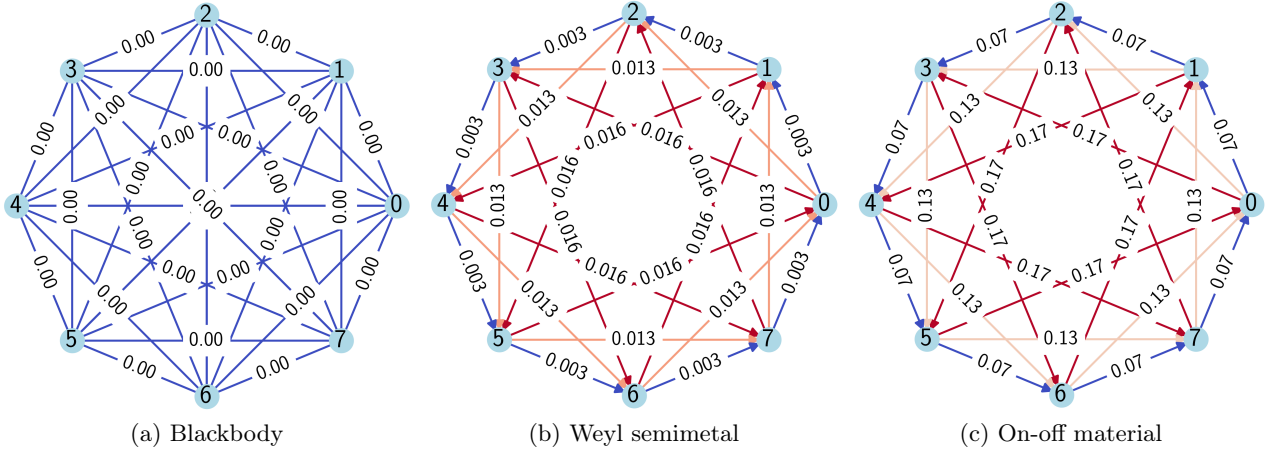


Figure 5: Graph representation of the monochromatic net transmission coefficient, $H_{i \rightarrow j} = S_{i \rightarrow j} - S_{j \rightarrow i}$ between 8 elements. In plot (a), no net heat transmission is observed as the material is reciprocal. In cases (b) and (c), a net heat transmission coefficient is observed due to the non-reciprocal nature of the material, however, the net heat leaving and entering every element is zero, as required by second law. No arrow means the transmission coefficient is zero. According to Section 3.2.2, the net heat transfer between each pair of elements i and j can be computed by integrating over all frequencies.

Result 1. *At thermal equilibrium, non-reciprocal materials exhibit a non-zero monochromatic net heat transfer between elements, whereas for reciprocal materials it is zero.*

The simulation has been thoroughly validated by ensuring:

1. (Emission conservation:) At each surface point, the average emitted power equals the absorbed heat coming from all other elements as second law dictates.
2. (Blackbody check:) For an ideal blackbody, both the average emissivity and the monochromatic transmitted coefficient at each mesh cell equal 1. Integrating the directional emissivity over all angles and frequencies recovers Planck's law and validates the sentence of Eq. (17).
3. (Radial symmetry:) The transmission graph exhibits radial symmetry for each element pair, consistent with the radial symmetry of a single made cylindrical material.

The following characteristics are observed in the behaviour of the monochromatic transmission coefficient:

Properties 1. *(Monochromatic transmission coefficient)*

- (Reciprocity:) *If the material is reciprocal then for every evenly spaced partition $n \in \mathbb{N}$, $S_{i \rightarrow j} = S_{j \rightarrow i}$.*
- (Second law constraint:) *The heat absorbed is balanced with the heat emitted by each element as second law dictates: $\sum_j S_{i \rightarrow j} = \sum_j S_{j \rightarrow i} \quad \forall i \in \{1, 2, \dots, n\}$.*
- (Self-transmission:) *Heat reabsorbed by the same element vanishes as the partition becomes finer: $\lim_{n \rightarrow \infty} \max_i \{S_{ii}\} = 0$*
- (Homogenous material:) *If the cylinder is made of a single material, then radial symmetry leads to $S_{i \rightarrow j} = S_{i+1 \rightarrow j+1}$ for all $i, j \in \{1, 2, \dots, n\}$* ⁹

We introduce the heat transfer matrix H and we use it to quantify how non-reciprocal a material is via a proposed non-reciprocity factor ζ .

⁹In the modulo sense: $j := j \bmod n$, $i := i \bmod n$

Definition 2. We define (elementwise) the heat transfer matrix $H \in \mathbb{M}_{n \times n}(\mathbb{R})$ as:

$$H_{ij} = S_{i \rightarrow j} - S_{j \rightarrow i}$$

Properties 2. (Net transfer matrix)

- (Second law constraint:) The heat absorbed and emitted by each element must balance as second law dictates. Thus, the sum of net transmission coefficients for a given element is zero:

$$\sum_j H_{ij} = 0 \quad \forall i \in \{1, 2, \dots, n\} \quad (20)$$

Definition 3. We define the non-reciprocity factor ζ for a cylindrical structure as:

$$\zeta = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i,j}^n |H_{ij}|$$

where n is the number of elements.

The non-reciprocity factor ζ behaves as modified entrywise or ℓ_1 matrix norm. It quantifies deviations from reciprocity: for reciprocal materials, $H_{ij} = 0 \quad \forall(i, j)$, leading to $\zeta = 0$. In contrast, non-reciprocal materials yield non-zero entries in H , resulting in $\zeta \neq 0$.

Properties 3. (Non-reciprocity factor)

1. (Existence and uniqueness:) The non-reciprocity factor exist and is unique.
2. (Reciprocal:) For reciprocal materials $\zeta = 0$.
3. (Non-reciprocal:) For non-reciprocal materials $\zeta > 0$.

Property one follows by considering the sequence ζ_n with $n \in \mathbb{N}$. This sequence is bounded above by 2, based on blackbody properties, and is monotonically increasing. This monotonicity arises because, even for a fully non-reciprocal material, the self-transmission terms must satisfy $H_{ii} = 0$ for all $i \in \{1, \dots, n\}$. Thus, as the partition is refined (with more elements), the reciprocal self-contributions decrease, making ζ_n to increase.

Table 2 presents the computed values of ζ for the aforementioned materials. No emissivity profile has been found that exceeds the value $\zeta = 1$. However, an analytical validation of this upper bound has not been obtained.

Material:	Blackbody	Weyl semimetal	On-off material
Factor ζ :	0.00	0.07	0.99

Table 2: Non-reciprocity factor ζ for various materials computed using a $n = 100$ element discretization. A value near 1 for the on-off material indicates strong non-reciprocal behaviour. The Weyl semimetal exhibits partial non-reciprocity $\zeta = 0.07$, while reciprocal materials yield $\zeta = 0$, consistent with theory.

Monochromatic transmission coefficient for a multiple-material cylinder

Additional results for configurations composed of different materials are presented in Figure 6. Two key observations arise. First, in a setup containing three non-reciprocal elements alternating with three blackbody elements, shown in panel 6a, a non-zero net transmission coefficient persists due to $S_{0 \rightarrow 2} = S_{2 \rightarrow 4} = S_{4 \rightarrow 0}$, recovering the results of [3], detailed in Section 3.1. Second, when only one element is non-reciprocal and the others are blackbodies, as shown in panel 6b, its influence propagates throughout the entire system. This is reasonable, as the non-reciprocal element interacts with

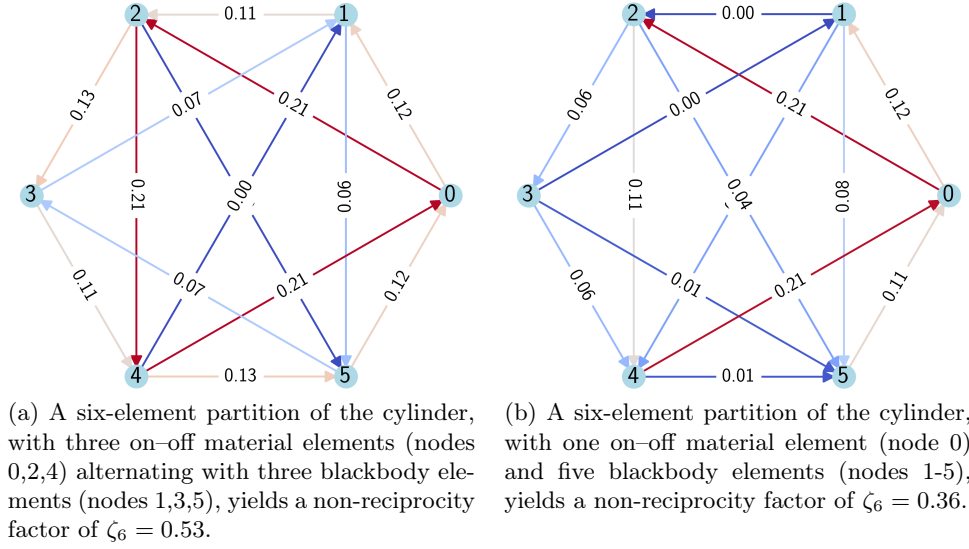


Figure 6: Graph representation of the net monochromatic transmission coefficient for a six-element multi-material partition of the cylinder.

all others via the ray-tracing algorithm.

Results from previous sections motivate to study the emergence of persistent heat currents within the cylinder. While non-zero transmission coefficients between elements indicate energy exchange, they do not necessarily imply a net directional heat flux, as energy transfer can also occur through multiple reflections in other directions.

4.2 Far-Field monochromatic inner and contour heat flux

Monochromatic heat flux vector fields in a single-material cylinder

The monochromatic heat flux vector fields inside and on the boundary of the cylinder are presented. We motivate a two dimensional plot by highlighting two fundamental properties of the resulting vector fields:

Properties 4. (Cylinder heat flux vector)

1. (Reciprocity in the z -direction:) Reciprocal emission profiles along the z -direction lead to heat flux vector fields $\mathbf{v}$ confined to the xy -plane. In other words, $\langle \mathbf{v}, (0, 0, 1) \rangle = 0 \forall \mathbf{v}$.
2. (Second law of thermodynamics:) At thermal equilibrium, the heat flux vector is tangential at the boundary of the cylinder. In other words, the vector field $\mathbf{v}$ satisfies $\langle \mathbf{v}, \mathbf{n} \rangle = 0$, where $\mathbf{n}$ is the outward normal vector.

The first property follows from the symmetry of the emissivity profile along the z -axis in Eq. (19). The second property is a consequence of the second law: if the heat flux was not tangential at the boundary, energy would enter or exit the closed surface, violating equilibrium. Moreover, since there are no internal sources within the cylinder, the heat flux vector field must be divergence-free [21].

In Figure 7, we present the plots for the three aforementioned materials described in Section 4.1. Persistent internal heat currents are observed only for the idealized on-off non-reciprocal material. For both the Weyl and reciprocal materials, internal currents vanish identically. We conclude that the behaviour observed in the on-off case is unphysical: zero emissivity creates isolated, vanishing heat channels that result in spurious persistent currents. Introducing a small non-zero emissivity reconnects these channels and vanishes internal currents. This effect is observed in the Weyl semimetals where the heat flux vanishes because its magnitude is equal in every direction. Although this might

appear unphysical at first glance, the relation $\alpha(\theta) = \epsilon(-\theta)$ enforces that. This occurs because the magnitude of the heat vector in any given direction results from both emission in that direction and rays coming from the opposite direction that are reflected into it. Due to the constraint $\epsilon(\theta) = \alpha(-\theta)$, strong emission implies that rays from the opposite direction are also strongly absorbed. Conversely, if the emission is weak in a given direction, then rays arriving from the opposite direction undergo weak absorption and are strongly reflected. Hence, the amplitude of the outgoing ray, which includes both reflection and emission, is relatively independent of the emissivity profile. As a consequence, the resulting magnitude of $\tilde{S}(\mathbf{p}, \theta, \varphi, \omega)$ in Eq. (19) remains constant in all directions, and the heat current vanishes.

On the other hand, non-reciprocal materials such as Weyl semimetals exhibit heat flux at the contour, even under thermal equilibrium, indicating preferential absorption from certain directions. This effect could provide useful insights for designing non-reciprocal thermal engines [10]. Such behaviour is absent in reciprocal materials, where no directional preference in heat exchange is observed. We underline that the contour heat flux is solely due to radiative heat transfer, since thermal conduction is neglected in the frame of this work.

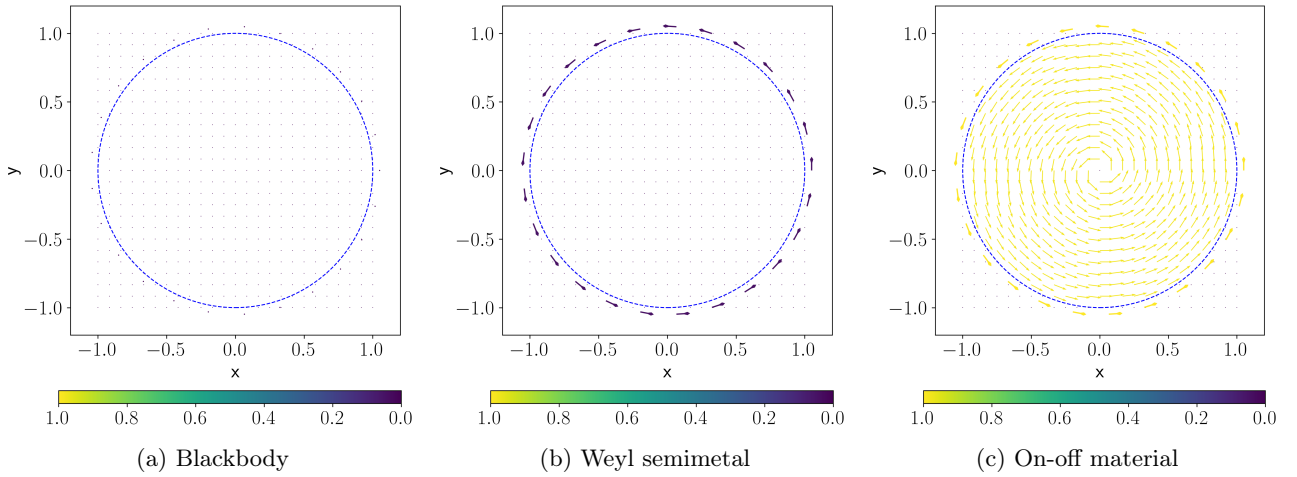


Figure 7: Inner and boundary heat flux vector fields on the xy -plane inside the cylinder. For the blackbody (a), no anomalous effects arise. For the Weyl semimetal (b), internal heat currents vanish, but a contour heat flux is observed. For the on-off material (c), an unphysical persistent heat current appears within the interior of the cylinder.

Result 2. *At thermal equilibrium, no persistent internal heat currents arise in physical reciprocal or non-reciprocal cylindrical configurations. However, non-reciprocal materials exhibit a directional contour heat flux, a behaviour absent in reciprocal systems.*

The simulation has been thoroughly validated by ensuring:

1. (Divergence-free heat flux:) The heat flux vector field is divergence-free, as given by ($v_\rho = v_z = 0$) and $\frac{\partial v_\theta}{\partial \theta} = 0$ validating [21]. Numerical results have been verified using a second-order Richardson extrapolation for approximating the derivatives.
2. (Tangential boundary heat flux:) Analytical verification showed that the heat flux vector field is tangential along the cylinder boundary at thermal equilibrium.

Monochromatic heat flux vector fields in a multiple-material cylinder and Non-equilibrium

Far-field heat currents are analysed for various material configurations to emulate a many-body system and to contrast our findings with the persistent heat currents arising in a triangular body configuration, as detailed in Section 3.1. We focus on combinations of reciprocal and non-reciprocal materials.

The blackbody is used to approximate vacuum conditions and on-off emission profiles are excluded due to their unphysical behaviour. Simulations include a six-body partition with alternating blackbody and non-reciprocal elements, emulating the triangular configuration studied in [3]. Additional tests were performed using mixed-material configurations composed of three and four elements of varying sizes. In all cases, our results show no persistent heat currents at thermal equilibrium.

This behaviour contrasts with the findings of Zhu and Fan’s three-body arrangement discussed in Section 3.1, primarily due to the different topology of our cylindrical configuration. In their case, persistent heat currents only arise at the boundaries of the three spheres and not at the center of the configuration, as shown in Figure 2. In our case, the cylindrical geometry can be interpreted as the limit case of a symmetric n -body configuration, where the number of bodies n tends to infinity. In this limit, the resulting heat current patterns resemble those shown in Figure 7b, with no heat currents in the inner part. Mathematically, we observe that the vanishing heat currents follow a reasoning similar to the single-material case and result from the directional balance imposed by the condition $\epsilon(\omega, \theta) = \alpha(\omega, -\theta)$, which holds independently of the specific material.

Result 3. *No persistent far-field heat currents arise inside the cylinder at thermal equilibrium for configurations composed of reciprocal and non-reciprocal physical materials.*

Finally, a **non-equilibrium** scenario is simulated. In Figure 8 we consider a six element configuration: three elements are modelled as Weyl semimetals, alternating with three idealized blackbodies at sufficiently low temperatures such that their emission is negligible. The resulting heat flux vector field exhibits a counterclockwise rotation, induced by the asymmetric enhancement of emissivity in the right hemisphere of the Weyl semimetal. The system is out of thermal equilibrium, as the heat flux is not tangential to the boundary. These results illustrate the potential for controlling heat flow under non-equilibrium conditions through appropriate material configurations.

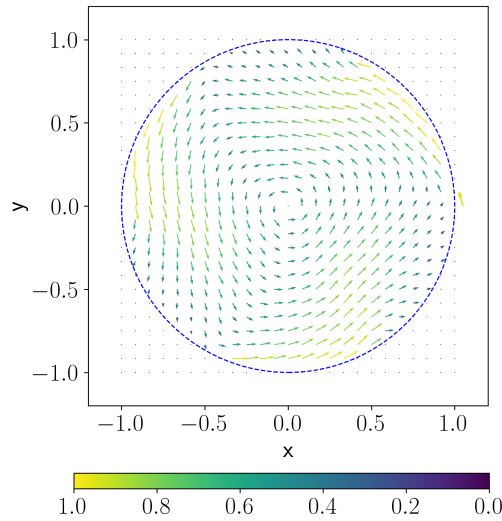


Figure 8: Heat flux vector field under non-thermal equilibrium conditions at the initial time. The cylinder is divided into six elements: three are modelled as Weyl semimetals, alternating with three idealised blackbodies at significantly lower temperatures. Directional heat currents emerge in a counterclockwise rotation, consistent with the asymmetric emissivity profile of the Weyl semimetals.

5 Final Remarks and Conclusion

This work investigated the consequences of breaking classical Kirchhoff’s law of thermal radiation for the first time in a non-planar cylindrical configuration, where analytical derivations for the transmission coefficient are unfeasible. An in-house ray-tracing algorithm was proposed and developed to study such scenarios. Our findings demonstrate that non-reciprocity can produce non-symmetric and non-zero net transmission coefficients between elements of the cylinder, thereby extending the results of Fan et al. [9] to non-planar surfaces. On the other hand, we show that no persistent far-field heat currents arise in the interior of a cylinder for various realistic compositions of reciprocal and non-reciprocal materials, contrasting with the results reported in [3] for the symmetric three-nanoparticle configuration. All our findings are consistent with the laws of thermodynamics and align with existing literature in both the reciprocal and non-reciprocal regimes. Finally, we note that no conclusions can be drawn for the class of colourless materials, as the specular ray-tracing algorithm is not suitable for modelling multiple diffractive channels [17].

Further improvements on the simulations could involve incorporating multiple diffractive channels into the ray-tracing algorithm, enabling the study of colourless materials. Additionally, extending the analysis to other geometries, such as a triangular configuration, remains an open and interesting problem. Experimental validation of non-reciprocal results is also necessary, since practical implementations often underperform compared to theoretical predictions. In such cases, optimizing material design and exploring broader wavelength ranges would further improve both theoretical insights and practical outcomes.

Overall, this thesis advances the understanding of non-reciprocal heat transfer in the far-field regime by presenting new results in a non-planar cylindrical cavity, opening new questions and directions for future research.

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A Fundamentals of Thermal Radiation

The aim of this section is to introduce the reader to the basic concepts of thermal radiation used in Section 2. The study of this field is an intersection between electromagnetism, optics and thermodynamics. The information of this section and notation will be mainly extracted from the book [22], additional concepts for the optics and photonics parts are extracted from [24], [25]. For the thermodynamics concepts we follow [26].

The **first law of thermodynamics** dictates that the energy of a system can be transferred with its surrounding via work and heat. Radiation is one of the three fundamental modes of heat transfer along with convection and conduction. The dual theory of the light explains the nature of radiation as a type of energy either emitted as electromagnetic waves or as a collection of photons when a body is at non-zero temperature. In this thesis, we focus on thermal radiation emitted by solid surfaces and how is interchanged between different materials ¹⁰.

The **second law of thermodynamics** dictates that the entropy of an isolated system never decreases and is maximum at thermal equilibrium. As a consequence, no net heat flow can be observed in an isolated system at thermal equilibrium since this would lead to an increase of entropy:

$$dS = \int \frac{\delta Q_{\text{rev}}}{T} = \frac{1}{T} \int \delta Q_{\text{rev}} > 0$$

The **third law of thermodynamics** imposes the existence of 0K temperature and affirms that in practice it is not possible to reach the 0K temperature. Consequently, every realistic body emits radiation.

A.1 Blackbody radiation and Planck's law

Every body at a non-zero temperature emits heat in the form of light. We define a blackbody as an idealized body perfectly opaque with zero reflectivity and a continuous spectrum of emission depending only on the wavelength and the temperature. According to Planck's distribution law, the spectral energy density of a black-body is given by:

$$u_{\omega}(\omega, T) = \frac{h\omega^3}{\pi^2 c^3} \frac{1}{e^{\frac{h\omega}{k_B T}} - 1} = g(\omega, T)\Theta(\omega, T)$$

Where ω is the frequency, c the speed of light, k_B the Boltzman constant, $\hbar$ Planck's reduced constant and T the temperature. This formula is decomposed as a product of two functions:

$$g(\omega, T) = \frac{\omega^2}{\pi^2 c^3} \quad \text{and} \quad \Theta(\omega, T) = \frac{\hbar\omega}{e^{\frac{h\omega}{k_B T}} - 1}$$

Where $g(\omega, T)$ is the degeneration number for each frequency level and $\Theta(\omega, T)$ is the mean photon energy distribution. These two terms are obtained using the Bose-Einstein distribution and assuming that the discretized energy levels that a photon can occupy are the same as a quantum harmonic oscillator. Consequently, each level ω can be occupied by $g(\omega, T)$ possible photons, with a mean energy per photon of $\Theta(\omega, T)$. In some articles, see explanation in [27], it is also common to write the zero point energy of the quantum harmonic oscillator $\Omega(\omega, T)$ as $\Omega(\omega, T) + \frac{\hbar\omega}{2}$. This factor is usually neglected as it is always erased when computing the difference of heat between bodies.

¹⁰The study of thermal radiation is of major importance in many engineering applications ranging from combustion and high-temperature metals, space cooling and insulation. Three aspects differentiate radiation from conduction and convection. Firstly, it can transfer energy without a medium, secondly, it depends on the wavelength and temperature and, finally, it depends on the orientation of the body or surface.

When computing the irradiance, namely, the flux of photons that are emitted by a blackbody, one considers a gas of photons confined in a cavity and follows the formalism described in [28] to obtain Planck's law:

$$B_b(\omega, T) = \frac{\hbar\omega^3}{4\pi^3c^2} \frac{1}{e^{\hbar\omega/(k_B T)} - 1}$$

Where B_b is defined as the irradiance of a blackbody. The total emission of a Blackbody is then given by integrating overall directions and frequencies, leading to the Stefan-Boltzmann law:

$$P = \int_0^\infty d\omega \int_0^{\pi/2} \int_0^{2\pi} B_b(\omega, T) \cos(\theta) \sin(\theta) d\theta d\varphi = \sigma T^4$$

Where $\sigma = \frac{2k_B^4\pi^5}{15c^2h^3}$ is known as the Stefan-Boltzmann constant.

A.2 Emission, absorption, reflection and transmission coefficients by Real Surfaces

The blackbody is an ideal emitter in the sense that, at a fixed temperature, no surface can emit more radiation than a blackbody. For these reason the blackbody is chosen as a reference in describing emissions of a real bodies. We define the spectral directional emissivity $\epsilon(\omega, \theta, \varphi, T)$ as the ratio between the spectral radiation emitted by the material and by the blackbody at the same values for a fixed T and ω :

$$\epsilon(\omega, \theta, \varphi, T) = \frac{B_{\text{material}}(\omega, \varphi, \theta, T)}{B_b(\omega, T)}$$

We define the *spectral emissivity*¹¹ as:

$$\epsilon(\omega, T) = \frac{\int_0^{2\pi} \int_0^{\pi/2} B_{\text{material}}(\omega, \theta, \varphi, T) \cos(\theta) \sin(\theta) d\theta d\varphi}{\int_0^{2\pi} \int_0^{\pi/2} B_b(\omega, T) \cos(\theta) \sin(\theta) d\theta d\varphi}$$

And finally the overall emissivity as:

$$\epsilon(T) = \frac{\int_0^\infty \epsilon_\omega(\omega, T) B_b(\omega, T) d\omega}{B_b(T)}$$

It is also common, see [22], to compute the overall emissivity averaging over all angles, in that case, it is also possible to compute $\epsilon(\omega, T)$ by:

$$\epsilon(\omega, T) = \frac{1}{\pi} \int_0^{2\pi} \int_0^{\pi/2} \epsilon(\omega, \varphi, \theta, T) \cos(\theta) \sin(\theta) d\theta d\varphi$$

The spectral directional absorptivity $\alpha(\omega, \varphi, \theta, T)$ is defined following the exact previous procedure but comparing now the irradiation absorbed by the surface (B_{abs}) with the irradiation absorbed by a black body.

$$\alpha(\omega, \theta, \varphi, T) = \frac{B_{\text{abs, material}}(\omega, \varphi, \theta, T)}{B_b(\omega, T)}$$

And the spectral absorptivity as:

$$\alpha(\omega, T) = \frac{\int_0^{2\pi} \int_0^{\pi/2} B_{\text{abs, material}}(\omega, \theta, \varphi, T) \cos(\theta) \sin(\theta) d\theta d\varphi}{\int_0^{2\pi} \int_0^{\pi/2} B_b(\omega, T) \cos(\theta) \sin(\theta) d\theta d\varphi}$$

And finally the overall absorptivity as:

$$\alpha(T) = \frac{\int_0^\infty \alpha_\omega(\omega, T) B_b(\omega, T) d\omega}{B_b(T)}$$

¹¹The cosine is the so called Lambert's cosine law. See: [29]

Either from the conservation of energy or from the restriction of the second law of thermodynamics in thermal equilibrium, detailed in Appendix A, the total emissivity and absorptivity of a material must be equal:

$$\int \alpha(\theta, \varphi) \cos(\theta) \sin(\theta) d\theta d\varphi = \int \epsilon(\theta, \varphi) \cos(\theta) \sin(\theta) d\theta d\varphi$$

When considering the case where irradiation interacts with a semitransparent medium, two more variables are introduced. In such case, we suppose that a beam of light impinges a material and part of it gets reflected, absorbed and transmitted. We define the reflectivity (R) and the transmittivity (τ) as the proportion of the incidence light that is respectively reflected and transmitted. Similar to the previous definitions, we can define these values spectrally, and for every direction, see [22] for more details. The following restriction is always obtained:

$$\alpha(T) + R(T) + \tau(T) = 1$$

A.3 Maxwell's equations

Maxwell's equations are the fundamental tool for studying electrodynamics. Given the electric and magnetic field respectively termed by $\mathbf{E}(\mathbf{r}, t) \in \mathbb{C}^3$ and $\mathbf{H}(\mathbf{r}, t) \in \mathbb{C}^3$, the following equations hold:

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon} && \text{(Gauss's law)} \\ \nabla \cdot \mathbf{B} &= 0 && \text{(Gauss's law for magnetism)} \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} && \text{(Faraday's law)} \\ \nabla \times \mathbf{B} &= \mu \mathbf{J} + \mu \epsilon \frac{\partial \mathbf{E}}{\partial t} && \text{(Ampère-Maxwell law)} \end{aligned}$$

Where ρ is the electric charge density, and $\mathbf{J}$ is the electric current density. The constants ϵ and μ are the vacuum permittivity and permeability, respectively.

When dealing with magnetic mediums, inhomogeneous or isotropic, it is common to work the Maxwell's equations in other fields that may take into account the properties of the mediums. In such case, the constitutive relations are given by:

$$\begin{aligned} \mathbf{D} &= \overleftrightarrow{\epsilon} \mathbf{E} + \overleftrightarrow{\xi} \mathbf{H} \\ \mathbf{B} &= \overleftrightarrow{\mu} \mathbf{H} + \overleftrightarrow{\zeta} \mathbf{E} \\ \mathbf{J} &= \overleftrightarrow{\sigma} \mathbf{E} \end{aligned}$$

Where $\mathbf{D}$ is the electric displacement field, $\mathbf{B}$ is the magnetic field, $\mathbf{E}$ is the electric field, $\mathbf{H}$ is the magnetic field intensity, and $\mathbf{J}$ is the current density. The symbols $\epsilon, \mu, \zeta, \eta, \sigma$ are respectively the electric permittivity, magnetic permeability, electric-magnetic coupling strength, and magnetoelectric coupling strength and electrical conductivity of the medium.

In this thesis, we focus in the theory of heat transfer when ϵ is a complex non symmetric tensor.

A.4 Boundary conditions and Wave equation

Applying Stokes and Gauss theorems, Maxwell's equation also provide constraints in the boundaries. Given two mediums 1 and 2, with respective magnetic and electric fields, $\mathbf{E}, \mathbf{D}, \mathbf{H}, \mathbf{B}$ the following results hold in between the surfaces:

$$\begin{aligned} \text{Tangential component of } \mathbf{E} \text{ is continuous : } & \mathbf{E}_1^{\parallel} = \mathbf{E}_2^{\parallel} \\ \text{Normal component of } \mathbf{D} \text{ satisfies : } & \mathbf{D}_2^{\perp} - \mathbf{D}_1^{\perp} = \rho \\ \text{Tangential component of } \mathbf{H} \text{ satisfies : } & \mathbf{H}_2^{\parallel} - \mathbf{H}_1^{\parallel} = \mathbf{K}_s \times \mathbf{n} \\ \text{Normal component of } \mathbf{B} \text{ satisfies : } & \mathbf{B}_1^{\perp} = \mathbf{B}_2^{\perp} \end{aligned}$$

Where ρ is the surface charge density, $\mathbf{K}_s$ surface current density and $\mathbf{n}$ is the normal vector. These tangential constraints follow either from taking closed loops or small cylindrical volumes in between the surfaces and applying the previously mentioned theorems.

To solve Maxwell's equations, it is common to derive a wave equation from Faraday's law by taking the curl of both sides and then using the Ampère–Maxwell law to express the result in terms of the electric field $\mathbf{E}$:

$$\begin{aligned}\nabla \times (\nabla \times \mathbf{E}) &= -\nabla \times \frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times (\nabla \times \mathbf{E}) &= -\mu \frac{\partial \mathbf{J}}{\partial t} - \mu \varepsilon \frac{\partial^2 \mathbf{E}}{\partial t^2}\end{aligned}$$

Rearranging terms appropriately, we obtain the wave equation for the electric field $\mathbf{E}$:

$$\nabla \times (\nabla \times \mathbf{E}) - \mu \varepsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} - \mu g \frac{\partial \mathbf{E}}{\partial t} = 0$$

A similar result holds for the magnetic field $\mathbf{H}$:

$$\nabla \times (\nabla \times \mathbf{H}) - \mu \varepsilon \frac{\partial^2 \mathbf{H}}{\partial t^2} - \mu g \frac{\partial \mathbf{H}}{\partial t} = 0$$

Here, μ is the magnetic permeability, ε is the electric permittivity, and g represents an effective conductivity term. Using the wave equation when solving Maxwell's equations offers a useful approach for generalizing Fresnel's equations in anisotropic media, enabling the computation of absorption and transmission coefficients.

A.5 Lorentz reciprocity

We follow the derivation provided in [30] and [31] to introduce reciprocity. Consider a source a , characterized by the electric and magnetic current densities $\mathbf{J}_a$ and $\mathbf{M}_a$, creating an electromagnetic field $(\mathbf{E}_a, \mathbf{H}_a)$, interacting with the electromagnetic field $(\mathbf{E}_b, \mathbf{H}_b)$ produced by another source b with corresponding currents $\mathbf{J}_b$ and $\mathbf{M}_b$. The interaction between source a and the field of b is quantified by the bilinear form.

$$\langle a, b \rangle = \int_V dV (\mathbf{J}_a \cdot \mathbf{E}_b - \mathbf{M}_a \cdot \mathbf{H}_b)$$

Where V is the volume of both source a and source b .

A system is called **reciprocal** with respect to the sources a and b if:

$$\langle a, b \rangle = \langle b, a \rangle$$

The previous formula states that the source and the detector of electromagnetic fields can be interchanged without affecting the physical situation of the problem. For example, given an isotropic material, e.g. in free space, the interaction between two points where one is the source and the other is the detector can be interchanged and the reaction will be the same. However, in anisotropic media, where certain directions have privilege over others, changing the location of the emitter and the receiver can change the strength or type of interaction.

We now present the necessary conditions for a material to hold reciprocity. Consider Maxwell's equations for the fields generated by source a and b given by:

$$\begin{aligned}\nabla \times \mathbf{H}_a &= -i\omega \mathbf{D}_a + \mathbf{J}_a & \nabla \times \mathbf{H}_b &= -i\omega \mathbf{D}_b + \mathbf{J}_b \\ \nabla \times \mathbf{E}_a &= i\omega \mathbf{B}_a + \mathbf{M}_a & \nabla \times \mathbf{E}_b &= i\omega \mathbf{B}_b + \mathbf{M}_b\end{aligned}$$

Take the following vector identity for two vector fields:

$$\nabla \cdot (\mathbf{E}_b \times \mathbf{H}_a) = \mathbf{H}_a \cdot (\nabla \times \mathbf{E}_b) - \mathbf{E}_b \cdot (\nabla \times \mathbf{H}_a)$$

Using the previous derivation for both a and b , we obtain:

$$\begin{aligned} -\nabla \cdot (\mathbf{E}_b \times \mathbf{H}_a) &= -(\mathbf{E}_b \cdot (\nabla \times \mathbf{H}_a) - \mathbf{H}_a \cdot (\nabla \times \mathbf{E}_b)) \\ &= -i\omega (\mathbf{E}_a \cdot \mathbf{D}_b - \mathbf{H}_a \cdot \mathbf{B}_b) + \mathbf{J}_a \cdot \mathbf{E}_b - \mathbf{M}_b \cdot \mathbf{H}_a \end{aligned} \quad (25)$$

Similarly:

$$-\nabla \cdot (\mathbf{E}_a \times \mathbf{H}_b) = -i\omega (\mathbf{E}_b \cdot \mathbf{D}_a - \mathbf{H}_b \cdot \mathbf{B}_a) + \mathbf{J}_b \cdot \mathbf{E}_a - \mathbf{M}_a \cdot \mathbf{H}_b \quad (26)$$

We subtract Eq. (25) from Eq. (26) and we integrate over a sufficiently large volume V , in such case, the left hand side expression is translated to a closed integral surface using Gauss theorem which vanishes, leading to:

$$\oint_S dS \cdot (\mathbf{E}_a \times \mathbf{H}_b - \mathbf{E}_b \times \mathbf{H}_a) = 0$$

Hence:

$$\langle a, b \rangle - \langle b, a \rangle = i\omega \int_V (\mathbf{E}_b \cdot \mathbf{D}_a - \mathbf{E}_a \cdot \mathbf{D}_b + \mathbf{H}_a \cdot \mathbf{B}_b - \mathbf{H}_b \cdot \mathbf{B}_a) dV$$

Using the general linear constitutive relations (see Section A.3):

$$\begin{aligned} \mathbf{D} &= \overleftrightarrow{\epsilon} \mathbf{E} + \overleftrightarrow{\xi} \mathbf{H} \\ \mathbf{B} &= \overleftrightarrow{\mu} \mathbf{H} + \overleftrightarrow{\zeta} \mathbf{E} \end{aligned}$$

we substitute into the expression to obtain

$$\begin{aligned} \langle a, b \rangle - \langle b, a \rangle &= i\omega \int_V [\mathbf{E}_b \cdot (\overleftrightarrow{\epsilon} - \overleftrightarrow{\epsilon}^T) \mathbf{E}_a - \mathbf{H}_a \cdot (\overleftrightarrow{\mu} - \overleftrightarrow{\mu}^T) \mathbf{H}_b \\ &\quad + \mathbf{E}_a \cdot (\overleftrightarrow{\xi} + \overleftrightarrow{\xi}^T) \mathbf{H}_b - \mathbf{H}_b \cdot (\overleftrightarrow{\zeta} + \overleftrightarrow{\zeta}^T) \mathbf{E}_a] dV \end{aligned}$$

Therefore, the necessary and sufficient conditions for the medium to be reciprocal are:

$$\begin{aligned} \overleftrightarrow{\epsilon} &= \overleftrightarrow{\epsilon}^T \\ \overleftrightarrow{\mu} &= \overleftrightarrow{\mu}^T \\ \overleftrightarrow{\xi} &= -\overleftrightarrow{\zeta}^T \end{aligned}$$

This leads to the generalized reciprocity theorem, which in the presence of a complementary system can be stated as:

$$\langle b, a \rangle = \langle a, b \rangle^C$$

Where the upperscript C refers to the interaction in the complementary system, defined in Section 2.2.

A.6 Dyadic Green's Formula

In electromagnetic theory, the dyadic Green's function $\overleftrightarrow{\mathbf{G}}(\mathbf{r}, \mathbf{r}')$ provides a convenient way of solving Maxwell's equations by using the superposition principle. Heuristically, it represents the electric field $\mathbf{E}(\mathbf{r})$ at an observation point $\mathbf{r}$ due to a point electric dipole source $\mathbf{p}$ located at $\mathbf{r}'$:

$$\mathbf{E}(\mathbf{r}) = \omega^2 \mu_0 \mu \overleftrightarrow{\mathbf{G}}(\mathbf{r}, \mathbf{r}') \mathbf{p}$$

Here, $\overleftrightarrow{\mathbf{G}}(\mathbf{r}, \mathbf{r}')$ is a rank-2 tensor, typically represented as a 3×3 matrix. It is historically called a *dyadic* to emphasize the linear mapping between vector source distributions and resulting fields.

To determine $\overleftrightarrow{\mathbf{G}}(\mathbf{r}, \mathbf{r}')$, one solves the inhomogeneous wave equation for a point source as detailed in Section A.4:

$$\nabla \times \nabla \times \overleftrightarrow{\mathbf{G}}(\mathbf{r}, \mathbf{r}') - k_0^2 \varepsilon \overleftrightarrow{\mathbf{G}}(\mathbf{r}, \mathbf{r}') = \mathbf{I} \delta(\mathbf{r} - \mathbf{r}')$$

Given the Green's function, the total electric field from an arbitrary current distribution $\mathbf{j}(\mathbf{r}')$ can be expressed as:

$$\mathbf{E}(\mathbf{r}) = \mathbf{E}_0(\mathbf{r}) + i\omega\mu_0\mu \int_V \overleftrightarrow{\mathbf{G}}(\mathbf{r}, \mathbf{r}') \mathbf{j}(\mathbf{r}') dV', \quad \mathbf{r} \notin V$$

This integral represents the field at point $\mathbf{r}$ as a superposition of infinitesimal source contributions at positions $\mathbf{r}'$. In homogeneous, isotropic free space, the dyadic Green's function takes the form:

$$\overleftrightarrow{\mathbf{G}}(\mathbf{r}, \mathbf{r}') = \left(\overleftrightarrow{\mathbf{I}} + \frac{1}{k^2} \nabla^2 \right) \frac{e^{ik|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|}, \quad k = \omega\sqrt{\mu\varepsilon}$$

Reciprocity. The concept of a complementary (or adjoint) system arises in the context of electromagnetic reciprocity. In such systems, the Green's function satisfies the reciprocity relation:

$$\overleftrightarrow{\mathbf{G}}(\mathbf{r}, \mathbf{r}') = \left[\overleftrightarrow{\mathbf{G}}^C(\mathbf{r}', \mathbf{r}) \right]^T$$

where $\overleftrightarrow{\mathbf{G}}^C$ is the Green's function of the adjoint system (see Section 2.2), and T denotes the transpose. This relation follows from the Green's function reciprocity theorem, derived from the identity:

$$\int_{\partial V} (\mathbf{E} \times \mathbf{H}^C - \mathbf{E}^C \times \mathbf{H}) \cdot d\mathbf{S} = \int_V (\mathbf{E}^C \cdot \mathbf{J} - \mathbf{E} \cdot \mathbf{J}^C) dV$$

The proof of this identity is standard and can be found in texts such as [25]. As previously shown, this general reciprocity principle yields:

$$0 = \langle a, b \rangle - \langle b, a \rangle^C = \int_{\partial V} (\mathbf{E} \times \mathbf{H}^C - \mathbf{E}^C \times \mathbf{H}) \cdot d\mathbf{S} = \int_V (\mathbf{E}^C \cdot \mathbf{J} - \mathbf{E} \cdot \mathbf{J}^C) dV$$

This implies:

$$\int_V \mathbf{E} \cdot \mathbf{J}^C dV = \int_V \mathbf{E}^C \cdot \mathbf{J} dV \quad (28)$$

Now, consider two point dipoles with current densities:

$$\begin{aligned} \mathbf{J}^C(\mathbf{r}) &= -i\omega\mathbf{p} \delta(\mathbf{r} - \mathbf{r}') \\ \mathbf{J}(\mathbf{r}) &= -i\omega\mathbf{q} \delta(\mathbf{r} - \mathbf{r}) \end{aligned}$$

Substituting into Eq. (28), we obtain:

$$\mathbf{q} \cdot \mathbf{E}(\mathbf{r}) = \mathbf{p} \cdot \mathbf{E}^C(\mathbf{r}')$$

Using the Green's function definitions:

$$\begin{aligned} \mathbf{E}(\mathbf{r}) &= \omega^2 \mu_0 \mu \overleftrightarrow{\mathbf{G}}(\mathbf{r}, \mathbf{r}') \mathbf{p} \\ \mathbf{E}^C(\mathbf{r}') &= \omega^2 \mu_0 \mu \overleftrightarrow{\mathbf{G}}^C(\mathbf{r}', \mathbf{r}) \mathbf{q} \end{aligned}$$

This yields:

$$\mathbf{q} \cdot \overleftrightarrow{\mathbf{G}}(\mathbf{r}, \mathbf{r}') \mathbf{p} = \mathbf{p} \cdot \overleftrightarrow{\mathbf{G}}^C(\mathbf{r}', \mathbf{r}) \mathbf{q}$$

Since this must hold for arbitrary vectors $\mathbf{p}$ and $\mathbf{q}$, it follows that:

$$\overleftrightarrow{\mathbf{G}}^T(\mathbf{r}, \mathbf{r}') = \overleftrightarrow{\mathbf{G}}^C(\mathbf{r}', \mathbf{r})$$

Implication for the reflectivity operator This relation implies:

$$\langle \omega, \hat{n}', \sigma' | \mathbf{R} | \omega, -\hat{n}, \sigma \rangle = \langle \omega, \hat{n}, \sigma | \tilde{\mathbf{R}} | \omega, -\hat{n}', \sigma' \rangle \quad (30)$$

Here, the notation follows Section 2.1, and $\sigma \in \{s, p\}$ denotes polarization states. For general polarization states:

$$|\hat{p}\rangle = \sum_{\sigma} c_{\sigma} |\sigma\rangle, \quad |\hat{p}'\rangle = \sum_{\sigma'} d_{\sigma'} |\sigma'\rangle$$

Then:

$$\begin{aligned} \langle \omega, \hat{n}', \hat{p}' | \mathbf{R} | \omega, -\hat{n}, \hat{p} \rangle &= \sum_{\sigma} \sum_{\sigma'} d_{\sigma'}^* c_{\sigma} \langle \omega, \hat{n}', \sigma' | \mathbf{R} | \omega, -\hat{n}, \sigma \rangle \\ &= \sum_{\sigma} \sum_{\sigma'} c_{\sigma} d_{\sigma'}^* \langle \omega, \hat{n}, \sigma | \tilde{\mathbf{R}} | \omega, -\hat{n}', \sigma' \rangle \\ &= \langle \omega, \hat{n}, \hat{p}^* | \tilde{\mathbf{R}} | \omega, -\hat{n}', \hat{p}'^* \rangle \end{aligned}$$

A.7 Symmetric emissivity profile and photonic band structures

The photonic band structure amplifies the notion of the limit angle in Snell's law and provides the conditions for total internal reflection in a given material. When analyzing wave transmission into a medium, we consider not only the propagating modes but also the evanescent waves. All the information is encoded in the photonic band structure $\omega(\mathbf{k}_{\parallel})$ which provides the allowed transmitted frequencies given an in-plane wavevector. Heuristically, the values for $\omega(\mathbf{k}_{\parallel})$ are given by a characteristic equation which depends strongly on the out-of-plane wave numbers k_{z1} and k_{z2} , see [25]. In the context of our work, it is important to note that the photonic band structure is simplified significantly due to the fact that the refractive index of vacuum is lower than that of the cylinder. As a result, all directions support transmission, and no total internal reflection occurs.

For reciprocal materials, we showed that $\overleftrightarrow{\epsilon} = \overleftrightarrow{\epsilon}^T$, and, according to the spectral theorem, there exists a basis in which the permittivity tensor becomes diagonal. In such cases, the material exhibits an ellipse diffraction index, meaning that the values of the permittivity tensor are invariant under opposite direction. As a consequence, the photonic band structure is symmetric in $\mathbf{k}_{\parallel}$ -space, i.e., $\omega(\mathbf{k}_{\parallel}) = \omega(-\mathbf{k}_{\parallel})$. This implies that the resonance frequencies for light incident at opposite angles are identical, and thus the angular-dependent emissivity satisfies $\epsilon(\omega, \theta, \varphi) = \epsilon(\omega, -\theta, \varphi)$.

For non-reciprocal materials, the permittivity tensor is no longer symmetric. This leads to an asymmetric photonic band structure, i.e., $\omega(k_x) \neq \omega(-k_x)$, and consequently, the emissivity can differ depending on the direction of incidence.

A.8 Poynting vector theorem:

By combining Maxwell's equations introduced in Section A.3, specifically, Faraday's law, the Ampère-Maxwell law, and the constitutive relations for $\mathbf{B}$ and $\mathbf{D}$, we arrive at the following expression:

$$\mathbf{H} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{H}) = -\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} - \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} - \mathbf{j} \cdot \mathbf{E}$$

Now using the properties for the divergence operator and applying Gauss's theorem over a large volume V the above equation becomes:

$$\int_{\partial V} (\mathbf{E} \times \mathbf{H}) \cdot \mathbf{n} dS = - \int_V \left[\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} + \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} + \mathbf{j} \cdot \mathbf{E} \right] dV$$

The right-hand side of the previous equation is interpreted as the variation of the energy inside a volume V and the left-hand side is interpreted as the energy leaving or entering volume V through

the boundary ∂V . This relation is called the Poynting vector theorem, and leads to the definition of Poynting vector:

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

In this sense, $\mathbf{S}$ represent the flow of energy of the system.

A.9 Fourier transforms

Given a real-valued signal $A(t)$, we define the Fourier transform using the following convention:

$$A(t) = \int_{-\infty}^{\infty} A(\omega) e^{i\omega t} d\omega = \text{Re} \left[\int_0^{\infty} A(\omega) e^{i\omega t} d\omega \right]$$

The second equality is true due to the fact that for a real valued signal $A(\omega) = A^*(-\omega)$.

We further define an spatial fourier transform from position to parallel wavevectors because We also define a spatial Fourier transform from position to transverse (parallel) wavevectors. This is useful for two reasons: first, the magnitude of the transverse wavevector determines whether the wave is propagating or evanescent; second, it encodes the angle of incidence, which is crucial when applying the Fresnel equations. We adopt the following convention:

$$A(\mathbf{r}_{\parallel}, z) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{d\mathbf{k}_{\parallel}}{(2\pi)^2} A(\mathbf{k}_{\parallel}, z) e^{-i\mathbf{k}_{\parallel} \cdot \mathbf{r}_{\parallel}}$$

B Preliminars on Non-reciprocal heat transfer theory

B.1 Transfer Matrix method

In this section, we introduce the transfer matrix method, used in Section 2.2, as the tool to obtain the reflectivity and transmitted matrices $\mathbf{R}$ and $\mathbf{T}$ of a material. The theory is obtained from the *The Transfer-Matrix Method in Electromagnetics and Optics* book, see [14].

The transfer-matrix method (TMM) in electromagnetism and optics is a convenient mathematical formalism for determining the reflection and transmission coefficients of incidence light when the material is anisotropic and Fresnel's equations are no longer possible to apply¹². Suppose we have a planar slab of finite thickness along the z -axis, homogeneous in the xy -plane. We derive an expression relating the incoming fields $\mathbf{E}$ and $\mathbf{H}$ to the outgoing fields $\mathbf{E}'$ and $\mathbf{H}'$. To this end, we restrict the analysis to planar waves and express the electric and magnetic fields as:

$$\begin{aligned}\mathbf{E}(r) &= (e_x(z)\mathbf{x} + e_y(z)\mathbf{y} + e_z(z)\mathbf{z})e^{i(\mathbf{k}\cdot\mathbf{r}+\omega t)} \\ \mathbf{H}(r) &= (h_x(z)\mathbf{x} + h_y(z)\mathbf{y} + h_z(z)\mathbf{z})e^{i(\mathbf{k}\cdot\mathbf{r}+\omega t)}\end{aligned}$$

Using the constitutive relations and Maxwell's equations (Appendix A.3), and applying the boundary conditions appropriately (Appendix A.4), one obtains a coupled system of differential equations for the variables e_x, e_y, h_x, h_z . Here, $z = 0$ denotes the field at the surface, and z corresponds to a point inside the slab:

$$\frac{d}{dz} \begin{bmatrix} e_x(z) \\ e_y(z) \\ h_x(z) \\ h_y(z) \end{bmatrix} = i\hat{\mathbf{P}} \begin{bmatrix} e_x(0) \\ e_y(0) \\ h_x(0) \\ h_y(0) \end{bmatrix}$$

The 4×4 matrix $\hat{\mathbf{P}}$ depends on the material's dielectric (ϵ) and magnetic (μ) permittivities, the angle of incidence, and the frequency ω . For simplicity, we omit its explicit form, however, a detailed expression is available in [14], Section 2.5.1. As $\hat{\mathbf{P}}$ is independent of z , the system's solution can be expressed analytically as:

$$\begin{bmatrix} e_x(z) \\ e_y(z) \\ h_x(z) \\ h_y(z) \end{bmatrix} = \hat{\mathbf{M}} \begin{bmatrix} e_x(0) \\ e_y(0) \\ h_x(0) \\ h_y(0) \end{bmatrix} \quad \hat{\mathbf{M}} = e^{i\hat{\mathbf{P}}z}$$

It is common to express the fields in terms of (s) and (p) polarizations. Due to Maxwell's equations, the electric field $\mathbf{E}$ and magnetic field $\mathbf{H}$ are interrelated, so it suffices to consider only $\mathbf{E}$. The total field can be decomposed into incident, reflected, and transmitted components:

$$\begin{aligned}\mathbf{E} &= \mathbf{E}_{\text{inc}} + \mathbf{E}_{\text{ref}} + \mathbf{E}_{\text{tr}} \\ \mathbf{E}_{\text{inc}}(r) &= (a_s\mathbf{E}^{(s)} + a_p\mathbf{E}^{(p)})e^{i(\mathbf{k}\cdot\mathbf{r}+\omega t)} \\ \mathbf{E}_{\text{ref}}(r) &= (r_s\mathbf{E}^{(s)} + r_p\mathbf{E}^{(p)})e^{i(\mathbf{k}\cdot\mathbf{r}+\omega t)} \\ \mathbf{E}_{\text{tr}}(r) &= (t_s\mathbf{E}^{(s)} + t_p\mathbf{E}^{(p)})e^{i(\mathbf{k}\cdot\mathbf{r}+\omega t)}\end{aligned}$$

To express the fields in the (s) and (p) polarization basis, it is common to apply a change of basis using two 4×4 matrices, $\hat{\mathbf{K}}_{\text{inc}}$ and $\hat{\mathbf{K}}_{\text{tr}}$. These matrices depend exclusively on the angles of incidence and transmission and the refractive indices of the incoming and outgoing media.

$$\begin{bmatrix} e_x(0) \\ e_y(0) \\ h_x(0) \\ h_y(0) \end{bmatrix} = \hat{\mathbf{K}}_{\text{inc}} \begin{bmatrix} a_s \\ a_p \\ r_s \\ r_p \end{bmatrix} \quad \begin{bmatrix} e_x(z) \\ e_y(z) \\ h_x(z) \\ h_y(z) \end{bmatrix} = \hat{\mathbf{K}}_{\text{tr}} \begin{bmatrix} 0 \\ 0 \\ t_s \\ t_p \end{bmatrix}$$

¹²The transfer matrix method formalism applies to locally planar smooth surfaces.

Combining the previous equations with the matrix $\hat{\mathbf{M}}$, one obtains the **transfer matrix** that relates the values of the incoming and outgoing field.

$$\begin{bmatrix} t_s \\ t_p \\ 0 \\ 0 \end{bmatrix} = \underbrace{\hat{\mathbf{K}}_{\text{tr}}^{-1} \cdot \hat{\mathbf{M}} \cdot \hat{\mathbf{K}}_{\text{inc}}}_{\text{transfer matrix}} \cdot \begin{bmatrix} a_s \\ a_p \\ r_s \\ r_p \end{bmatrix}$$

Finally, by algebraically manipulating the previous equations, it is possible to define the transmitted and reflectivity matrices, that relate the amplitude factors of the reflected and transmitted field:

$$\begin{bmatrix} r_s \\ r_p \end{bmatrix} = \begin{bmatrix} r_{ss} & r_{sp} \\ r_{ps} & r_{pp} \end{bmatrix} \cdot \begin{bmatrix} a_s \\ a_p \end{bmatrix} \quad \begin{bmatrix} t_s \\ t_p \end{bmatrix} = \begin{bmatrix} t_{ss} & t_{sp} \\ t_{ps} & t_{pp} \end{bmatrix} \cdot \begin{bmatrix} a_s \\ a_p \end{bmatrix}$$

The reflectivity and transmitted matrices $\mathbf{R}$ and $\mathbf{T}$ are then defined as:

$$\mathbf{R} = \begin{bmatrix} r_{ss} & r_{sp} \\ r_{ps} & r_{pp} \end{bmatrix} \quad \mathbf{T} = \begin{bmatrix} t_{ss} & t_{sp} \\ t_{ps} & t_{pp} \end{bmatrix}$$

The coefficients r_{ij}, t_{ij} are complex-valued and describe the relation between the amplitude of the incoming field in polarization $i \in \{s, p\}$ and the amplitude of the reflected or transmitted field in polarization $j \in \{s, p\}$.

A Julia package implementing the transfer matrix method is currently under development, see: `GeneralizedTransferMatrixMethod.jl`.

B.2 Emissivity and absorption profiles via the Transfer Matrix method

We present the theoretical framework, used in section 2.2, for defining the emissivity and absorptivity matrices of a material using the transfer matrix formalism. The approach follows the methodology outlined in [15].

Consider a linear object O that receives an incident input field $|\phi_{\text{in}}\rangle$ and produces an output field $|\phi_{\text{out}}\rangle$. The interaction between the input and output fields is described by the scattering matrix, which is an algebraic transformation of the transfer matrix (see Section B.1). This matrix plays a role analogous to the reflectivity and transmissivity matrices $\mathbf{R}$ and $\mathbf{T}$, but it directly relates incoming and outgoing fields.

$$|\phi_{\text{out}}\rangle = \mathbf{S}|\phi_{\text{in}}\rangle$$

For the following discussion, we may neglect the specific structure and dimensionality of the scattering matrix $\mathbf{S}$, we only apply the singular value decomposition¹³ to $\mathbf{S}$:

$$\mathbf{S} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\dagger = \sum_{p \in \mathcal{P}} s_p |\phi_{\text{out}-p}\rangle \langle \phi_{\text{in}-p}|$$

Here $|\phi_{\text{in}-p}\rangle$ and $|\phi_{\text{out}-p}\rangle$ denote properly normalized input and output mode eigen-vectors with singular value associated s_p (eigenvalue of the diagonal matrix $\mathbf{\Sigma}$). The set of modes of the system is referred as $(\mathcal{P})$.

A procedure to work out the problem for each mode p is presented. To do so, we adopt the formalism of *single-mode blackbodies*, where the object O is assumed to be completely surrounded by ideal blackbodies, each of which emits and receives only a single mode from a set $\{|\phi_{\text{Qin}-p}\rangle, |\phi_{\text{Qout}-p}\rangle\}_{p \in \mathcal{P}}$. This theoretical construction has been shown to be valid in [32]. These new modal functions are related to the original input and output modes via unitary transformations $\hat{\mathbf{U}}$ and $\hat{\mathbf{V}}$, such that

$$|\phi_{\text{Qin}-p}\rangle = \hat{\mathbf{U}} |\phi_{\text{in}-p}\rangle, \quad |\phi_{\text{Qout}-p}\rangle = \hat{\mathbf{V}} |\phi_{\text{out}-p}\rangle, \quad \forall p \in \mathcal{P}$$

¹³The singular value decomposition (SVD) decomposes a matrix into a product of two unitary matrices and a diagonal matrix.

For each blackbody mode $|\phi_{Q_{in-p}}\rangle$, the absorptivity at mode p is directly computed as the fraction of total incident power that is absorbed:

$$\alpha_p = 1 - |\mathbf{S}|\phi_{Q_{in-p}}\rangle|^2 = 1 - \langle\phi_{Q_{in-p}}|\mathbf{S}^\dagger\mathbf{S}|\phi_{Q_{in-p}}\rangle$$

If we take into account the linearity of the material, we can define the **absorptivity matrix** as:

$$\mathcal{A} = \mathbf{I} - \mathbf{S}^\dagger\mathbf{S} \quad (\text{Absorptivity matrix})$$

By linearity, for any input state $|\phi\rangle$, the absorption coefficient is given by

$$\alpha = \langle\phi|\mathcal{A}|\phi\rangle,$$

where $\mathcal{A}$ denotes the absorption operator.

The emissivity in channel p of a blackbody consists of the direct emission from the object O into channel p , along with contributions from all other channels q that, after interacting with O , scatter into channel p . By definition of the scattering matrix $\mathbf{S}$, the amplitude scattered into port p from an input in port q is given by:

$$S_{pq} := \langle\phi_{Q_{out-p}}|\mathbf{S}|\phi_{Q_{in-q}}\rangle.$$

Therefore, the total scattered power from all the other modes to mode p is:

$$P_B \sum_{q \in \mathcal{P}} |S_{pq}|^2 = P_B \sum_{q \in \mathcal{P}} \langle\phi_{Q_{out-p}}|\mathbf{S}|\phi_{Q_{in-q}}\rangle \langle\phi_{Q_{in-q}}|\mathbf{S}^\dagger|\phi_{Q_{out-p}}\rangle = P_B \langle\phi_{Q_{out-p}}|\mathbf{S}\mathbf{S}^\dagger|\phi_{Q_{out-p}}\rangle$$

Here, P_B is the irradiance of a blackbody and we use the completeness relation:

$$\sum_{q \in \mathcal{P}} |\phi_{Q_{in-q}}\rangle \langle\phi_{Q_{in-q}}| = \mathbf{I}$$

The total power emitted by the object into channel p is the sum of the object's intrinsic emission (ϵP_B) in channel p and the scattered contributions from all other modes:

$$Q_{out-p} = \epsilon P_B + \langle\phi_{Q_{out-p}}|\mathbf{S}\mathbf{S}^\dagger|\phi_{Q_{out-p}}\rangle P_B$$

Finally, since we are in thermal equilibrium, the total emitted power into channel p is recieved by the blackbody at channel p and balanced out by its emission, which is B_p , leading to:

$$P_B = \epsilon P_B + \langle\phi_{Q_{out-p}}|\mathbf{S}\mathbf{S}^\dagger|\phi_{Q_{out-p}}\rangle P_B \implies \epsilon = 1 - \langle\phi_{Q_{out-p}}|\mathbf{S}\mathbf{S}^\dagger|\phi_{Q_{out-p}}\rangle$$

Leading to the definition of the **emissivity matrix** as:

$$\mathcal{E} = \mathbf{I} - \mathbf{S}\mathbf{S}^\dagger$$

If the material is opaque, the matrix $\mathbf{S}$ reduces to the reflectivity matrix $\mathbf{R}$ detailed in Section B.1.

B.3 Equalities in Heat transfer coefficient formulation section

We unfold the chain of equalities in Section 3.2.1, Eq.(14).

Equality 1 Eq. (14a): For equality 1, it is useful to first compute the Fourier transform of the Poynting vector and study the function as a sum of planar waves, we apply the Fourier transform convention defined in section A.9:

$$\begin{aligned} \langle\mathbf{S}(\mathbf{r}, t)\rangle &= \langle\mathbf{E}(\mathbf{r}, t) \times \mathbf{H}(\mathbf{r}, t)\rangle \\ &= \text{Re} \left\langle \left(\int_0^\infty \hat{\mathbf{E}}(\mathbf{r}, \omega) e^{i\omega t} d\omega \right) \times \left(\int_0^\infty \hat{\mathbf{H}}(\mathbf{r}, \omega') e^{i\omega' t} d\omega' \right) \right\rangle \\ &\stackrel{(*)}{=} \text{Re} \int_0^\infty \int_0^\infty \langle \hat{\mathbf{E}}(\mathbf{r}, \omega) e^{i\omega t} \times \hat{\mathbf{H}}(\mathbf{r}, \omega') e^{i\omega' t} \rangle \delta(\omega - \omega') d\omega d\omega' \\ &= \text{Re} \int_0^\infty \langle \hat{\mathbf{E}}(\mathbf{r}, \omega) e^{i\omega t} \times \hat{\mathbf{H}}(\mathbf{r}, \omega) e^{i\omega t} \rangle d\omega \end{aligned}$$

Where in (*) we use that there is no correlation between fields of different frequencies.

The integrand in the previous equation can be written more efficiently as:

$$\begin{aligned}
 \text{Re} \left\langle \hat{\mathbf{E}}(\mathbf{r}, \omega) e^{i\omega t} \times \hat{\mathbf{H}}(\mathbf{r}, \omega) e^{i\omega t} \right\rangle &= \frac{1}{T} \int_0^T \text{Re}[\mathbf{E} e^{i\omega t}] \times \text{Re}[\mathbf{H} e^{i\omega t}] dt \\
 &= \frac{1}{4T} \int_0^T [\mathbf{E} e^{i\omega t} + \mathbf{E}^* e^{-i\omega t}] \times [\mathbf{H} e^{i\omega t} + \mathbf{H}^* e^{-i\omega t}] dt \\
 &= \frac{1}{4T} \int_0^T [\mathbf{E} \times \mathbf{H}^* + \mathbf{E}^* \times \mathbf{H} + \mathbf{E} \times \mathbf{H} e^{2i\omega t} + \mathbf{E}^* \times \mathbf{H}^* e^{-2i\omega t}] dt \\
 &= \frac{1}{4T} \int_0^T [\mathbf{E} \times \mathbf{H}^* + \mathbf{E}^* \times \mathbf{H}] dt \\
 &= \frac{1}{2} \text{Re} [\mathbf{E}(\mathbf{r}, \omega) \times \mathbf{H}^*(\mathbf{r}, \omega)]
 \end{aligned}$$

When studying the heat transmitted, it is useful to study the Poynting vector in terms of the parallel wavevector $\mathbf{k}_{\parallel}$ as it provides information about the angle of incidence, which is essential for computing the flux. Moreover, the value of the parallel wavevector provides whether the wave is propagating or evanescent¹⁴. Consider the use of local coordinates: For every point $p = [\alpha, z]$ in the cylinder, we define the local coordinate system given by:

$$\begin{aligned}
 \mathbf{e}_1 &= \frac{1}{R} \frac{\partial \varphi}{\partial \alpha} = (-\sin(\alpha), \cos(\alpha), 0) \\
 \mathbf{e}_2 &= \frac{\partial \varphi}{\partial z} = (0, 0, 1) \\
 \mathbf{e}_3 &= (\cos(\alpha), \sin(\alpha))
 \end{aligned}$$

We write $\mathbf{r} = x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3 = (\mathbf{r}_{\parallel}, z)$ so that z represent the direction normal to the cylinder at point p . Now we perform the spatial Fourier transform in this local coordinate system $(x, y) = \mathbf{r}_{\parallel} \rightarrow \mathbf{k}_{\parallel}$. For this step, we also suppose that the electric and magnetic fields at different space points are uncorrelated:

$$\begin{aligned}
 \langle \mathbf{S}(\mathbf{r}, \omega) \rangle &= \frac{1}{2} \text{Re} \int_0^{\infty} [\mathbf{E}(\mathbf{r}, \omega) \times \mathbf{H}^*(\mathbf{r}, \omega)] d\omega \\
 &= \frac{1}{2} \text{Re} \int_0^{\infty} d\omega \int \int \frac{d\mathbf{k}'_{\parallel} d\mathbf{k}_{\parallel}}{(2\pi)^4} [\mathbf{E}(\mathbf{k}_{\parallel}, \omega) \times \mathbf{H}^*(\mathbf{k}'_{\parallel}, \omega)] \\
 &= \frac{1}{2} \text{Re} \int_0^{\infty} d\omega \int \int \frac{d\mathbf{k}'_{\parallel} d\mathbf{k}_{\parallel}}{(2\pi)^4} [\mathbf{E}(\mathbf{k}_{\parallel}, z, \omega) \times \mathbf{H}^*(\mathbf{k}'_{\parallel}, z, \omega)] \delta(\mathbf{k}_{\parallel} - \mathbf{k}'_{\parallel}) \\
 &= \frac{1}{2} \text{Re} \int_0^{\infty} d\omega \int \frac{d\mathbf{k}_{\parallel}}{(2\pi)^4} [\mathbf{E}(\mathbf{k}_{\parallel}, z, \omega) \times \mathbf{H}^*(\mathbf{k}_{\parallel}, z, \omega)]
 \end{aligned}$$

The final expression for the Poynting vector is:

$$\langle \mathbf{S}(\mathbf{r}, \omega) \rangle = \frac{1}{2} \text{Re} \int_0^{\infty} d\omega \int \frac{d\mathbf{k}_{\parallel}}{(2\pi)^4} [\mathbf{E}(\mathbf{k}_{\parallel}, z, \omega) \times \mathbf{H}^*(\mathbf{k}_{\parallel}, z, \omega)]$$

The expression for the ensemble averaged value of the Poynting vector in terms of the wave vector enables us to compute the heat that is transmitted to the surface. It is assumed that the surface of the cylinder is smooth enough so that we can define a normal vector at every point $\mathbf{r} \in \Omega_i$. Then the heat transmitted will be given by the flux crossing the surface or equivalently:

$$\langle \mathbf{S}(\mathbf{r}, t) \cdot \mathbf{n} \rangle = \frac{1}{2} \text{Re} \int_0^{\infty} d\omega \int \frac{d\mathbf{k}_{\parallel}}{(2\pi)^4} \mathbf{n} \cdot [\mathbf{E}(\mathbf{k}_{\parallel}, z, \omega) \times \mathbf{H}^*(\mathbf{k}_{\parallel}, z, \omega)] \quad (36)$$

¹⁴In this thesis we only focus on propagating waves, which are the only ones persisting in far field heat transfer. Evanescent waves decay exponentially with distance, see [25].

Equality 2: Eq. (14b) The goal of equality 2 is to derive a better expression for the cross product of Eq. (36). We can carry out the study for planar waves each one with a fixed frequency ω and direction $\mathbf{k}_{\parallel}$ ¹⁵.

We decompose the electric and magnetic field in the (s) perpendicular and (p) parallel to the plane of incidence polarizations:

$$\mathbf{E} = \mathbf{E}^{(s)} + \mathbf{E}^{(p)}$$

This uncouples the cross product into (s) and (p) and we can study the behaviour of each polarizations separately, similar to Fresnel's equations:

$$\mathbf{E} \times \mathbf{H} = (\mathbf{E}^{(s)} + \mathbf{E}^{(p)}) \times (\mathbf{H}^{(s)} + \mathbf{H}^{(p)}) = \mathbf{E}^{(s)} \times \mathbf{H}^{(s)} + \mathbf{E}^{(p)} \times \mathbf{H}^{(p)}$$

$$\text{Since } \mathbf{E}^{(p)} \times \mathbf{H}^{(s)} = \mathbf{E}^{(s)} \times \mathbf{H}^{(p)} = 0$$

We carry out the study only with the (s) polarization, the other one is identical. The goal is to obtain $\mathbf{E}^{(s)}$ and $\mathbf{H}^{(s)}$ in order to perform the cross product in Eq. (36). Consider a planar wave $\mathbf{E}_i = \mathbf{y}E_i e^{ik_{1z}z + ik_{1\parallel}x}$. When the incident field strikes the surface, it is partially reflected and partially transmitted, each with different intensities. The magnetic field exhibits a similar behaviour and is coupled to the electric field through Maxwell's equations.

$$\mathbf{E}^{(s)} = \begin{cases} \mathbf{E}_i^{(s)} = \mathbf{y}E_i e^{ik_{iz}z + ik_{i\parallel}x} \\ \mathbf{E}_r^{(s)} = \mathbf{y}E_r e^{ik_{rz}z + ik_{r\parallel}x} \\ \mathbf{E}_t^{(s)} = \mathbf{y}E_t e^{ik_{tz}z + ik_{t\parallel}x} \end{cases} \quad \mathbf{H}^{(s)} = \begin{cases} \mathbf{H}_i^{(s)} = \left(\frac{-k_z}{w\mu_m} E_i e^{ik_{iz}z + ik_{i\parallel}x}, 0, \frac{k_x}{w\mu_m} E_i e^{ik_{iz}z + ik_{i\parallel}x} \right) \\ \mathbf{H}_r^{(s)} = \left(\frac{-k_z}{w\mu_m} E_r e^{ik_{rz}z + ik_{r\parallel}x}, 0, \frac{k_x}{w\mu_m} E_r e^{ik_{rz}z + ik_{r\parallel}x} \right) \\ \mathbf{H}_t^{(s)} = \left(\frac{-k_z}{w\mu_m} E_t e^{ik_{tz}z + ik_{t\parallel}x}, 0, \frac{k_x}{w\mu_m} E_t e^{ik_{tz}z + ik_{t\parallel}x} \right) \end{cases}$$

Where we have used that for planar waves:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \implies \mathbf{k} \times \mathbf{E} = \omega \mu_m \mathbf{H}$$

Now we impose the boundary conditions of the Maxwell's equations. Using the Stokes theorem on a closed loop crossing the surface with an area that goes to zero we obtain that the tangential electric field must be equal in both sides, leading to:

$$\mathbf{u}_n \times \mathbf{E}_i + \mathbf{u}_n \times \mathbf{E}_r = \mathbf{u}_n \times \mathbf{E}_t$$

Where $\mathbf{u}_n$ is the vector normal to the surface.

Since this dependence has to be satisfied for any position and time. This means that the three waves need to have the same functional dependence:

$$(\mathbf{k}_i \cdot \mathbf{r} - w_i t)|_{z=0} = (\mathbf{k}_r \cdot \mathbf{r} - w_r t + \epsilon_r)|_{z=0} = (\mathbf{k}_t \cdot \mathbf{r} - w_t t + \epsilon_t)|_{z=0}$$

And following the same argument as in the deduction of Fresnel's equations we obtain the the law of reflection and Snell's law that provide the values of the z-component of the transmitted and reflected wavevector:

$$\begin{aligned} \mathbf{k}_i \parallel &= \mathbf{k}_r \parallel = \mathbf{k}_t \parallel & \omega_i &= \omega_r = \omega_t \\ \mathbf{k}_i &= (\mathbf{k}_i \parallel, k_{1z}) & \mathbf{k}_r &= (\mathbf{k}_i \parallel, -k_{1z}) & \mathbf{k}_t &= (\mathbf{k}_i \parallel, k_{tz}) & \frac{\mathbf{k}_i \parallel}{k_{tz}} &= \frac{n_1}{n_2} \sin(\theta_1) \end{aligned}$$

With all this information, the components of the electric field in the boundary are computed. Yielding:

¹⁵Note that given a ω and $\mathbf{k}_{\parallel}$, one can infer the direction of the wave since $\mathbf{k}_z = \sqrt{\mathbf{k}^2 - \mathbf{k}_{\parallel}^2}$ and $\mathbf{k} = \frac{w}{c} \sqrt{\epsilon}$

$$\mathbf{E}^{(s)} = \mathbf{y}E_y = \begin{cases} (E_i e^{ik_{iz}z} + E_r e^{-ik_{iz}z}) e^{ik_{\parallel}x} & \text{for } z > 0 \\ E_t e^{ik_{tz}z} e^{ik_{\parallel}x} & \text{for } z < 0 \end{cases}$$

$$\mathbf{H}^{(s)} = (H_x, 0, H_z) = \begin{cases} H_x = \begin{cases} -\frac{k_{iz}}{\omega\mu_m} (E_i e^{ik_{iz}z} - E_r e^{-ik_{iz}z}) e^{ik_{\parallel}x} & \text{for } z > 0 \\ -\frac{k_{tz}}{\omega\mu_m} E_t e^{ik_{tz}z} e^{ik_{\parallel}x} & \text{for } z < 0 \end{cases} \\ H_z = \begin{cases} \frac{k_{\parallel}}{\omega\mu_m} (E_i e^{ik_{iz}z} + E_r e^{-ik_{iz}z}) e^{ik_{\parallel}x} & \text{for } z > 0 \\ \frac{k_{\parallel}}{\omega\mu_m} E_t e^{ik_{tz}z} e^{ik_{\parallel}x} & \text{for } z < 0 \end{cases} \end{cases}$$

And the desired value of the cross product is:

$$\mathbf{E}^{(s)} \times \mathbf{H}^{*(s)} = \begin{vmatrix} \mathbf{x} & \mathbf{y} & \mathbf{z} \\ 0 & E_y^{(s)} & 0 \\ H_x^{*(s)} & 0 & H_z^{*(s)} \end{vmatrix} = (E_y^{(s)} H_z^{*(s)}, 0, -H_x^{*(s)} E_y^{(s)})$$

For our computation, we are interested in the normal component incident to the surface, in our coordinate system, this is the z-component at $z = 0$. The magnetic and electric fields are then:

$$\mathbf{E}^{(s)} = \mathbf{y}E_y = \begin{cases} (E_i + E_r) e^{ik_{\parallel}x} & \text{for } z \searrow 0 \\ E_t e^{ik_{\parallel}x} & \text{for } z \nearrow 0 \end{cases}$$

$$\mathbf{H}^{(s)} = (H_x, 0, H_z) = \begin{cases} H_x = \begin{cases} -\frac{k_{iz}}{\omega\mu_m} (E_i - E_r) e^{ik_{\parallel}x} & \text{for } z \searrow 0 \\ -\frac{k_{tz}}{\omega\mu_m} E_t e^{ik_{\parallel}x} & \text{for } z \nearrow 0 \end{cases} \\ H_z = \begin{cases} \frac{k_{\parallel}}{\omega\mu_m} (E_i + E_r) e^{ik_{\parallel}x} & \text{for } z \searrow 0 \\ \frac{k_{\parallel}}{\omega\mu_m} E_t e^{ik_{\parallel}x} & \text{for } z \nearrow 0 \end{cases} \end{cases}$$

And we obtain that the value of the cross product is:

$$\mathbf{n} \cdot \mathbf{E}^{(s)}(\mathbf{k}_{\parallel}, z=0, \omega) \times \mathbf{H}^{*(s)}(\mathbf{k}_{\parallel}, z=0, \omega) = \frac{k_{iz}^*}{\omega\mu_m} (E_i + E_r)(E_i^* - E_r^*)$$

This is the result obtained for s-polarization. The result for both polarizations is the sum also for the p-polarization case. A compact form of writing this is using the trace, see [12]:

$$\begin{aligned} \mathbf{n} \cdot \mathbf{E}(\mathbf{k}_{\parallel}, 0, \omega) \times \mathbf{H}(\mathbf{k}_{\parallel}, 0, \omega) &= \mathbf{n} \cdot (\mathbf{E}^{(s)}(\mathbf{k}_{\parallel}, 0, \omega) \times \mathbf{H}^{*(s)}(\mathbf{k}_{\parallel}, 0, \omega) \\ &\quad + \mathbf{E}^{(p)}(\mathbf{k}_{\parallel}, 0, \omega) \times \mathbf{H}^{*(p)}(\mathbf{k}_{\parallel}, 0, \omega)) \\ &= \frac{k_{iz}^{*(s)}}{\omega\mu_m} (E_i^{(s)} + E_r^{(s)})(E_i^{*(s)} - E_r^{*(s)}) \\ &\quad + \frac{k_{iz}^{*(p)}}{\omega\mu_m} (E_i^{(p)} + E_r^{(p)})(E_i^{*(p)} - E_r^{*(p)}) \\ &= \text{Tr} \left[(\mathbf{E}_i + \mathbf{E}_r) [Z^{-1}(\mathbf{E}_i - \mathbf{E}_r)]^{\dagger} \right], \end{aligned}$$

Where $\mathbf{Z}$ is the 2×2 impedance matrix, see [22], and $\mathbf{E}_i$ and $\mathbf{E}_r$ are matrix and vectors given by:

$$\mathbf{Z} = \begin{pmatrix} Z^{(s)} & 0 \\ 0 & Z^{(p)} \end{pmatrix} \quad \mathbf{E}_i = \begin{pmatrix} \mathbf{E}_i^{(s)} \\ \mathbf{E}_i^{(p)} \end{pmatrix} \quad \mathbf{E}_r = \begin{pmatrix} \mathbf{E}_r^{(s)} \\ \mathbf{E}_r^{(p)} \end{pmatrix}$$

Equality 3: Eq. (14c) In Eq.(14c), the relation between the reflected and incidence fields will be given by the reflectivity matrix $\mathbf{R}(\theta)$ defined in section B.1:

$$\mathbf{E}_r = \mathbf{R}(\theta)\mathbf{E}_i$$

Making use of this identity we rewrite the relation in our previous result:

$$\begin{aligned} \mathbf{n} \cdot \mathbf{E}(\mathbf{k}\|, 0, \omega) \times \mathbf{H}^*(\mathbf{k}\|, 0, \omega) &= \text{Tr} \left[(\mathbf{E}_i + \mathbf{E}_r) [Z^{-1}(\mathbf{E}_i - \mathbf{E}_r)]^\dagger \right] \\ &= \text{Tr} \left[(\mathbf{E}_i + \mathbf{R}\mathbf{E}_i) [Z^{-1}(\mathbf{E}_i - \mathbf{R}\mathbf{E}_i)]^\dagger \right] \\ &= \text{Tr} \left[(\mathbf{1} + \mathbf{R})\mathbf{E}_i [Z^{-1}(\mathbf{1} - \mathbf{R})\mathbf{E}_i]^\dagger \right] \\ &= \text{Tr} \left[(\mathbf{1} + \mathbf{R})\mathbf{E}_i [Z^{-1}(\mathbf{1} - \mathbf{R})\mathbf{E}_i]^\dagger \right] \end{aligned}$$

The inside product is unfolded as:

$$\begin{aligned} (\mathbf{1} + \mathbf{R})\mathbf{E}_i [Z^{-1}(\mathbf{1} - \mathbf{R})\mathbf{E}_i]^\dagger &= Z^{-1}(\mathbf{1} + \mathbf{R})\mathbf{E}_i\mathbf{E}_i^\dagger(\mathbf{1} - \mathbf{R}^\dagger) \\ &= Z^{-1}(\mathbf{E}_i\mathbf{E}_i^\dagger + \mathbf{R}\mathbf{E}_i\mathbf{E}_i^\dagger - \mathbf{E}_i\mathbf{E}_i^\dagger\mathbf{R}^\dagger - \mathbf{R}\mathbf{E}_i\mathbf{E}_i^\dagger\mathbf{R}^\dagger) \\ &= Z^{-1}(\mathbf{E}_i\mathbf{E}_i^\dagger + \mathbf{R}\mathbf{E}_i\mathbf{E}_i^\dagger - (\mathbf{R}\mathbf{E}_i\mathbf{E}_i^\dagger)^\dagger - \mathbf{R}\mathbf{E}_i\mathbf{E}_i^\dagger\mathbf{R}^\dagger) \end{aligned}$$

Now let us use two properties. The first one being:

$$\text{Re}(\mathbf{R}\mathbf{E}_i\mathbf{E}_i^\dagger - (\mathbf{R}\mathbf{E}_i\mathbf{E}_i^\dagger)^\dagger) = 0$$

For the second identity we need to use the cyclic property of the trace:

$$\text{Tr}(\mathbf{ABC}) = \text{Tr}(\mathbf{BCA}) = \text{Tr}(\mathbf{CAB})$$

And as a consequence, we have that:

$$\text{Tr}(\mathbf{R}\mathbf{E}_i\mathbf{E}_i^\dagger\mathbf{R}^\dagger) = \text{Tr}(\mathbf{R}^\dagger\mathbf{R}\mathbf{E}_i\mathbf{E}_i^\dagger)$$

We finally obtain:

$$\text{Re } \mathbf{n} \cdot \mathbf{E}(\mathbf{k}\|, 0, \omega) \times \mathbf{H}^*(\mathbf{k}\|, 0, \omega) = \text{Tr} \left[(\mathbf{1} - \mathbf{R}^\dagger\mathbf{R})Z^{-1}\mathbf{E}_i\mathbf{E}_i^\dagger \right]$$

The final step is to compute the value of the forward field. The most common approach is to use the fluctuation-dissipation theorem. The magnitude of the emitted far-field by a body is given by:

$$\mathbf{E}_i\mathbf{E}_i^\dagger = (2\pi)^2 \frac{Z}{\pi} \Theta(\omega, T)(\mathbf{1} - \mathbf{R}\mathbf{R}^\dagger)$$

Where Θ is defined¹⁶ as:

$$\Theta = \frac{\hbar\omega}{e^{\frac{\hbar\omega}{k_b T}} - 1}$$

Note that we recover the formulas for the absorptivity and emissivity matrices provided in sec. A.2:

$$\mathcal{E} = \mathbf{I} - \mathbf{R}^\dagger\mathbf{R} \quad \mathcal{A} = \mathbf{I} - \mathbf{R}\mathbf{R}^\dagger$$

¹⁶It is also common to define Θ with an independent term called zero point energy, in such case, this independent term cancels out when computing net heat flux.

B.4 View factor between two surfaces

We provide the representation of the heat interchanged between two element mesh in 3.2.3. Given two surface elements dA_i and dA_j , the heat transmitted between them is defined as:

$$Q_{i \rightarrow j} = \frac{B_b(\omega, T) \cos(\theta_i) \cos(\theta_j)}{\pi S^2} \epsilon_i(\theta_i, \omega) \alpha_j(\theta_j, \omega) d\lambda dA_i dA_j$$

Figure 9 illustrates the interchange of heat denoted as Q_{ij} :

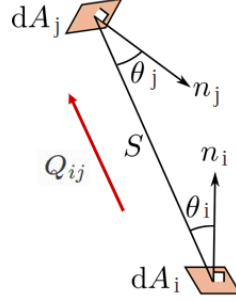


Figure 9: View factor between two non-parallel elements dA_i and dA_j

Here, θ_1 and θ_2 denote the angles between the normal of dA_1 and dA_2 , and the line connecting the two cells. S is the distance between mesh cells and $\epsilon_1(\theta_1, \omega)$, $\alpha_2(\theta_2, \omega)$ represent the emissivity and absorptivity profiles of mesh cells dA_1 , dA_2 respectively and $B_b(\omega, T)$ is the blackbody radiance.

Instead of integrating over surface mesh cells, we perform a change of variables to integrate over solid angles:

$$\frac{\cos(\theta_2)}{\pi S^2} dA_2 = d\Omega,$$

where $d\Omega$ is the differential of solid angle. This leads to:

$$dQ_{1 \rightarrow 2} = \int B(\lambda, T) \cos(\theta_1) \epsilon(\theta_1, \lambda) \alpha(\theta_2, \lambda) d\lambda dA_1 d\Omega$$

Thus, for each cell dA_1 rays are emitted uniformly in angular space. This reasoning shows that the distance S is not involved in the algorithm.

B.5 Colourless material Example

We introduce an example for a colourless material described in Section 2.3. The constitutive relations for a colourless material are given by:

$$\epsilon = \epsilon_0 \begin{pmatrix} 0.06 + 0.10i & 0.36 + 0.43i & 1.01 - 0.48i \\ 0.36 - 0.43i & 1.14 + 0.03i & 0.65 + 0.06i \\ 1.01 + 0.48i & 0.65 - 0.06i & 1.50 + 0.04i \end{pmatrix}$$

$$\mu = \mu_0 \begin{pmatrix} 1.41 + 0.01i & 0.82 + 0.45i & 0.70 - 0.08i \\ 0.82 - 0.45i & 1.73 + 0.08i & 0.54 - 0.21i \\ 0.70 + 0.08i & 0.54 + 0.21i & 0.59 + 0.054i \end{pmatrix}$$

$$\zeta = \eta^\dagger = \sqrt{\epsilon_0 \mu_0} \begin{pmatrix} 0.50 + 0.18i & 0.47 + 0.32i & 0.98 + 0.12i \\ 0.79 + 0.82i & 0.84 + 0.91i & 0.29 + 0.55i \\ 0.41 + 0.24i & 0.52 + 1.00i & 0.70 + 0.32i \end{pmatrix}$$

This example as well as the resulting emissivity and absorption profile is found in [2].

B.6 Fresnel's equations

Fresnel's equations describe the reflection and transmission of electromagnetic waves at the interface between two different homogeneous, isotropic media. These media are characterized by their electric permittivity ε and magnetic permeability μ . Consider a plane wave incident on a planar boundary separating medium 1 (with parameters ε_1, μ_1) and medium 2 (with parameters ε_2, μ_2). The interface lies in the xy -plane, and the wavevector components parallel to this interface are k_x and k_y . The components normal to the interface in each medium are denoted k_{z1} and k_{z2} , aligned with the z -axis.

One common form of defining Fresnel's equations is given by:

$$\begin{aligned} r_s(k_x, k_y) &= \frac{\mu_2 k_{z1} - \mu_1 k_{z2}}{\mu_2 k_{z1} + \mu_1 k_{z2}} & r_p(k_x, k_y) &= \frac{\varepsilon_2 k_{z1} - \varepsilon_1 k_{z2}}{\varepsilon_2 k_{z1} + \varepsilon_1 k_{z2}} \\ t_s(k_x, k_y) &= \frac{2\mu_2 k_{z1}}{\mu_2 k_{z1} + \mu_1 k_{z2}} & t_p(k_x, k_y) &= \frac{2\varepsilon_2 k_{z1}}{\varepsilon_2 k_{z1} + \varepsilon_1 k_{z2}} \sqrt{\frac{\mu_2 \varepsilon_1}{\mu_1 \varepsilon_2}} \end{aligned}$$

Here, the wavevector magnitudes and their components normal to the interface are computed as:

$$\begin{aligned} k_1 &= \frac{\omega}{c} \sqrt{\varepsilon_1 \mu_1} & k_{z1} &= \sqrt{k_1^2 - (k_x^2 + k_y^2)} \\ k_2 &= \frac{\omega}{c} \sqrt{\varepsilon_2 \mu_2} & k_{z2} &= \sqrt{k_2^2 - (k_x^2 + k_y^2)}. \end{aligned}$$

where ω is the angular frequency of the wave and c is the speed of light.

B.7 Codes

The datasets generated during and/or analyzed during the current study as well as the codes for the computations are available from the corresponding author on request.